

# Causal-Factored Cross-City Model Transfer for Bus Holding Control with Open Transit Data

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## Abstract

Hybrid offline-to-online reinforcement learning for transit control is attractive because operators often have rich historical data but only imperfect simulators. However, a bus holding policy tuned in one city may fail in another city unless the target simulator is manually calibrated, and such calibration is expensive. We study this cross-city transfer problem for bus holding using four open transit city bundles: Singapore, Austin/CapMetro, Halifax Transit, and MBTA. We propose Causal-Factored Cross-City Model Transfer (CFCMT), a mechanism-wise residual world-model adapter that decomposes dense simulator corrections into headway, demand/dwell, speed, and reward mechanisms, and selects source corrections using only unlabeled target-city simulator summaries. Against a dense  $H2O^+$ -style residual baseline, CFCMT is evaluated under a strict leave-one-city-out protocol in which each of the four cities is the target exactly once. On static-derived cross-city dynamics validation, unweighted CFCMT beats  $H2O^+$  on all four targets with mean total-MSE ratio 0.759, while similarity-weighted CFCMT also wins on all four targets and lowers the mean ratio to 0.578. Single-source zero-shot transfer is intentionally harder and mixed, with 6/12 wins and mean ratio 0.927, while few-shot target offline adaptation keeps the mechanism-factored model below the dense adapted baseline on most target-route budgets; the strongest few-shot variant is a low-dimensional target context-bias adapter rather than full mechanism refitting. In the full-module validation, a source-only feature-set selector plus guarded source-city ensemble lowers mean total MSE from 884.9 for dense  $H2O^+$ -style transfer and 851.9 for rigid CFCMT to 836.1 over six configured cross-city splits, while preserving 6/6 wins against  $H2O^+$ ; the guard uses unlabeled target summaries but no target transition labels. A same-information dense  $H2O^+$  source ensemble improves to 846.5 but remains weaker than the guarded mechanism-factored ensemble. In one-step policy-proxy evaluation, CFCMT improves reward/regret on three of four targets and lowers bunching rate on all four targets. Target-information, per-mechanism, ablation, route-bootstrap, source-subset, and generator-bias checks support the same main trend while showing that the advantage depends on multi-source diversity and should not be overstated as superiority on every small city subset. On two public Austin raw APC/AVL slices, source-only CFCMT is more stable than dense residual transfer but still worse than a state-informed passive no-correction baseline; with few-shot target residual calibration, a residual gate beats that passive baseline at 1% target line-group budget and at chronological budgets of six hours or more in both slices. Because passive bus logs cannot identify unobserved holding-action counterfactuals, we add a counterfactual traffic-signal-control stress test on RESCO. There, pressure-guarded CFCMT reaches MaxPressure-level mean queue on the 600-second multi-seed stress test and improves over dense residual transfer, simulator-only MPC, native actuated SUMO, and fixed signal programs, while a 3600-second check correctly exposes phase pressure as the stronger long-horizon classical rule. Full-budget official RESCO and patched LibSignal RL baselines are executed as native-protocol leaderboard cross-checks, not as same-protocol rows for the CFCMT phase-MPC table. The resulting contribution is a reproducible, reviewer-facing protocol and an empirically supported boundary condition: mechanism-factored transfer is useful for target-label-efficient cross-city residual adaptation, but counterfactual control or field validation remains necessary before operational claims.

# 1 Introduction

Bus holding control aims to reduce headway instability by delaying selected vehicles at stops. The control problem is local in execution but system-wide in consequence: a hold action changes forward and backward headways, passenger waiting, dwell, and downstream bunching. Modern reinforcement learning (RL) methods can learn holding policies in simulation, but practical deployment is constrained by two persistent gaps. First, high-fidelity simulation and manual calibration are expensive to reproduce in each city. Second, public data availability differs sharply across agencies: one city may expose speed bands and passenger OD data, while another may only provide GTFS schedules and coarse ridership summaries.

Hybrid offline-to-online methods such as  $H2O^+$  learn from both logged data and simulator interaction by correcting or reweighting dynamics gaps (Niu et al., 2022, 2025). This is a natural starting point for transit systems, where a historical buffer and a lower-fidelity simulator are often available. The difficulty is cross-city transfer. A dense monolithic residual model can fit simulator error in the source cities, but its parameters mix mechanisms that may shift differently across cities: schedule headways, route speeds, stop demand, and dwell behavior do not all change together. If cross-city transfer depends on localized changes in mechanisms, a factored representation should be more stable than a dense one (Bengio et al., 2020; Feng et al., 2022).

This paper studies that hypothesis in a deliberately conservative setting. We do not claim that the current data are vehicle-level real-world trajectories. Instead, we construct full-route open transit city bundles and evaluate models on static-data-derived offline dynamics targets. The protocol is meant to answer a narrower question: when only open static data and an uncalibrated simulator-like transition generator are available, does a mechanism-factored residual adapter transfer better across cities than a dense  $H2O^+$ -style residual correction? We separate three deployment regimes that are often conflated: single-source zero-shot transfer, multi-source target-label-free transfer, and few-shot target offline adaptation with a small amount of passive target-city history. This is the bus-holding core of the paper. Because passive bus logs cannot reveal what would have happened under unobserved hold actions, we use traffic-signal control as an additional counterfactual stress test where SUMO can roll out alternative actions under a shared benchmark protocol. The traffic-signal experiments are not presented as the same task or metric scale as bus holding; they test whether the same transfer principle can remain safe when strong classical control rules and external RL leaderboard baselines are available.

We introduce Causal-Factored Cross-City Model Transfer (CFCMT). CFCMT fits residual corrections at the level of output mechanisms rather than one dense residual over all outputs. For each target quantity, it uses a mechanism-specific sparse parent set: headway and gap outputs depend primarily on neighboring headways, target headway, speed, time, and action; demand and dwell outputs depend on waiting passengers, speed, target headway, time, and action; speed outputs use speed, time, and route-position features; reward uses policy-relevant headway and waiting quantities. We also add a target-label-free source-selection layer. It first chooses the target-summary feature set by leave-one-source validation, then uses the selected unlabeled target summary to decide when a source-city ensemble is credible; target transition labels are held out for evaluation. The theoretical motivation is a mechanism-risk decomposition: squared transition error decomposes over output mechanisms, source ensembles obey a convex risk bound under squared loss, and source-only selector quality depends on how well leave-one-source scores approximate target risk rankings.

The paper contributes:

1. A cross-city open-transit validation protocol using four full-route city bundles and a strict leave-

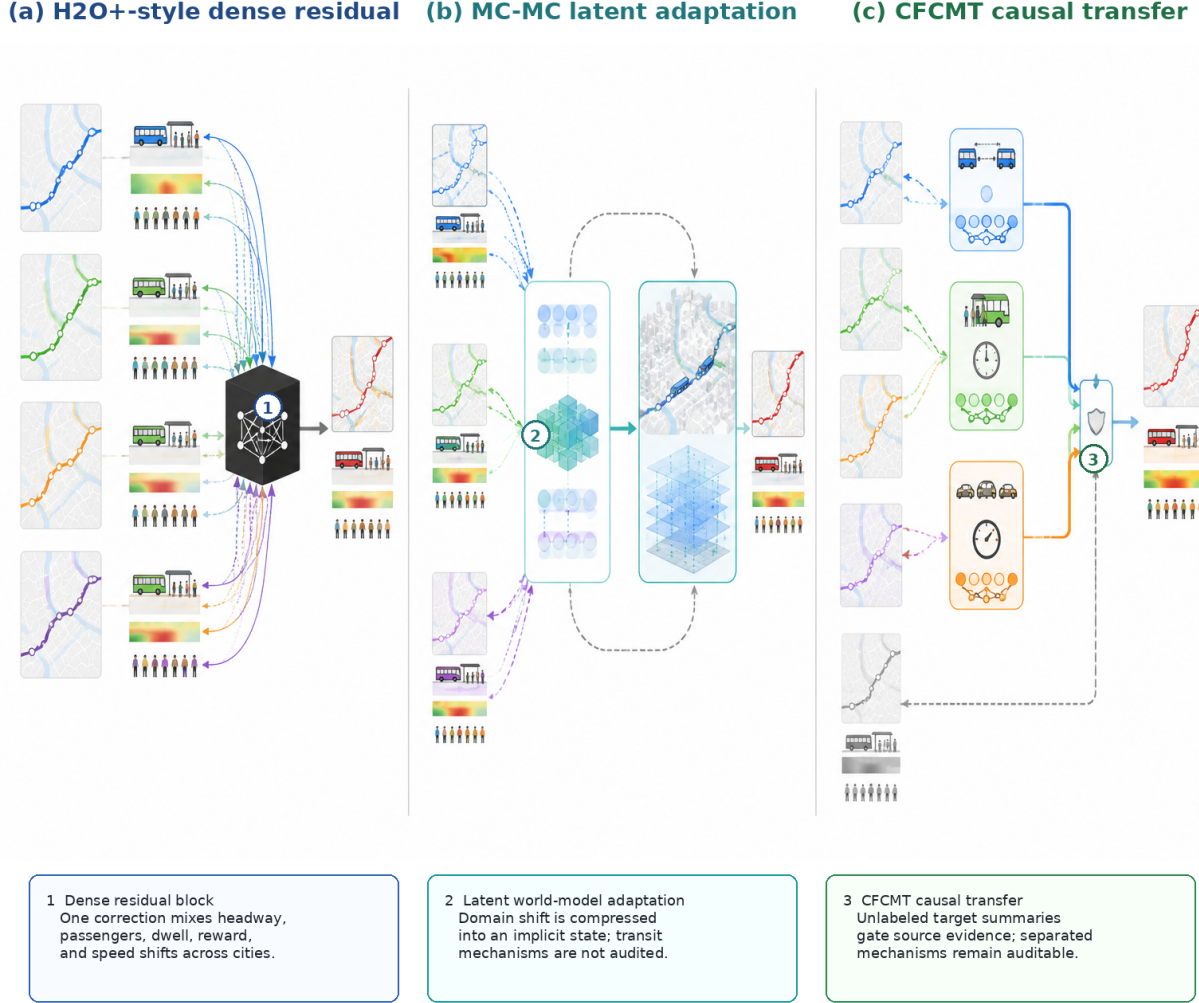


Figure 1: Conceptual comparison of cross-city residual transfer strategies for bus holding. A dense H2O<sup>+</sup>-style residual adapter learns one monolithic correction; MC-MC-style latent world-model adaptation represents domain shift implicitly; CFCMT instead transfers auditable headway/gap, demand/dwell, and speed mechanisms through a target-label-free source gate.

one-city-out evaluation unit.

2. A mechanism-wise residual transfer model with source-only target-summary feature selection and a guarded source-city ensemble.
3. A formal transfer-risk motivation showing how mechanism shift, source weighting, and source-only selector error enter the target risk.
4. A target-information ladder spanning single-source zero-shot transfer, multi-source target-label-free transfer, and few-shot target offline adaptation.
5. A counterfactual traffic-signal stress test showing that a pressure-guarded mechanism-transfer controller reaches pressure-rule-level safety on short RESCO rollouts, while long-horizon results favor a classical phase-pressure rule.

6. Full-budget native-protocol RESCO and LibSignal RL cross-checks, kept separate from the same-protocol CFCMT phase-MPC comparisons.
7. An explicit boundary on the claim: the evidence supports target-label-efficient cross-city residual transfer and safe-prior gating, not field deployment without counterfactual or vehicle-level validation.

## 2 Related Work

**Bus holding control.** Classical holding policies analyze headway stability and bunching through control rules and queueing approximations (Daganzo, 2009; Xuan et al., 2011). Recent work has applied deep RL and multi-agent RL to bus holding under stochastic demand and travel times (Alesiani and Gkiotsalitis, 2018; Wang and Sun, 2020; Liu et al., 2023; Wang and Sun, 2023). These studies typically focus on one calibrated network or one simulation benchmark. Our focus is different: we evaluate whether a learned dynamics correction transfers across city bundles with heterogeneous open data sources.

**Hybrid offline-to-online RL and dynamics gaps.**  $H2O^+$  and  $H2O^+$ -style methods address the mismatch between offline data and online simulator interaction by estimating dynamics-related weights or residuals (Niu et al., 2022, 2025). Related sim-to-real and offline-to-online methods use domain classifiers, calibrated Q-learning, residual dynamics, or world-model adaptation to reduce transfer error (Eysenbach et al., 2021; Nakamoto et al., 2023; Lanier et al., 2025). In this paper,  $H2O^+$  is treated as the dense residual-correction baseline: it can learn flexible source-city corrections, but the correction is not explicitly decomposed by transit mechanism.

**Causal and factored transfer.** The independent causal mechanisms principle suggests that distribution shifts may be localized when represented in appropriate mechanisms (Bengio et al., 2020). Factored adaptation methods use this idea to improve non-stationary RL and transfer by separating latent change factors or identifying the variables that need target-domain adaptation (Huang et al., 2022; Feng et al., 2022). CFCMT applies this principle to bus holding dynamics: rather than learning one dense correction, it learns mechanism-wise residuals aligned with interpretable transit outputs. The approach is intentionally lightweight in this paper: the reported cross-city experiments use sufficient-statistics ridge models for reproducibility, while the repository also contains neural causal-factored residual modules for later end-to-end training.

**Traffic simulation and open transit data.** SUMO provides a standard microscopic traffic simulation platform (Lopez et al., 2018). For cross-city public transit studies, however, fully calibrated SUMO networks and vehicle-level AVL/APC data are rarely available across many agencies. We therefore use open static data to create full-route city bundles. This makes the experiments reproducible but also limits their interpretation; the static-derived targets are not a substitute for live operational trajectory validation.

## 3 Problem Setting

Let  $s_t \in \mathcal{S}$  denote a bus-system state at a decision event and  $a_t \in \mathcal{A}$  denote a holding action in seconds. The state contains line and station identifiers, time of day, direction, forward and

backward headways, waiting passengers, target headway, base dwell, co-line headways, gap, and segment speed. The transition target is

$$\mathbf{y}_t = (h_{t+1}^+, h_{t+1}^-, q_{t+1}, d_{t+1}, g_{t+1}, \bar{h}_{t+1}^+, \bar{h}_{t+1}^-, v_{t+1}, r_{t+1}),$$

covering next headways, waiting passengers, dwell, gap, co-line headways, speed, and reward.

Each city  $c$  provides a bundle  $\mathcal{D}_c$  of route, timetable, speed, and demand tables. From these tables we construct paired simulator and static-derived target transitions:

$$\mathbf{y}^{\text{sim}} = f_0(s, a), \quad \mathbf{y}^{\text{target}} = f_c(s, a),$$

and train residual models on

$$\Delta_c(s, a) = \mathbf{y}^{\text{target}} - \mathbf{y}^{\text{sim}}.$$

For cross-city validation, source cities  $\mathcal{C}_{\text{src}}$  are used for fitting and a held-out target city  $c^*$  is used for evaluation. The target transition labels  $\mathbf{y}^{\text{target}}$  are never used for target-label-free model fitting or source selection. However, the target city’s unlabeled covariates, schedules, simulator outputs, speed summaries, and static local descriptors may be used to choose or weight source mechanisms.

**Target-information budgets.** We distinguish four information budgets. In *single-source zero-shot* transfer, one source city provides residual labels and the target city contributes only route/timetable/static-summary inputs. In *multi-source target-label-free* transfer, several source cities provide residual labels, while the target still provides no transition labels before reporting; unlabeled target summaries may be used for source weighting. In *few-shot target offline adaptation*, a small passive target history is available before deployment and may update mechanism trust, context bias, or residual sufficient statistics, with held-out target transitions reserved for evaluation. In *target-calibrated oracle* rows, the target budget is large or in-sample; these rows bound performance but are not no-calibration deployment evidence. The paper’s main claim concerns the second and third budgets, not pure one-city zero-shot deployment.

**Traffic-signal stress-test task.** The RESCO traffic-signal experiment is used as a counterfactual-control stress test rather than as the primary bus-holding task. For each signalized network  $n$ , each controlled traffic light  $i$ , and each decision time  $t$ , the state  $x_{n,i,t}$  contains phase-local queue, occupancy, waiting-time, and downstream-spillback summaries. The action  $a_{i,t}$  is one of the green phases already defined by the SUMO `tlLogic`; no synthetic phase set is introduced. The prediction target is the next local queue/cost summary after applying that phase for one control interval. Cross-network validation holds out one RESCO scenario as the target and fits phase-transfer models from the remaining scenarios. Zero-target-label rows use only source transitions plus target network descriptors and target-static simulator summaries; few-shot rows may use a short passive target slice to update a low-dimensional bias or trust parameter. Closed-loop target rollout metrics are used only for final evaluation.

The main evaluation unit is a city target, not a route. For four cities, the strict leave-one-city-out protocol has exactly four target evaluations:

$$\{(\mathcal{C} \setminus \{c\}) \rightarrow c : c \in \mathcal{C}\}.$$

This avoids counting repeated target cities as independent evidence. Additional configured split diagnostics are reported separately when they test source-selection behavior rather than the four-city claim.

**Claim scope.** The target transitions are static-data-derived pseudo-observations. They combine agency data, schedule-derived speeds or headways, passenger-demand summaries, and deterministic transition rules. They are useful for reproducible cross-city validation, but they are not vehicle-level AVL/APC ground truth. Accordingly, the paper evaluates transfer robustness under open static-data-derived dynamics, not operational field performance. Real passive APC/AVL slices validate observed action-0 dynamics and target offline adaptation, but they still do not provide labels for unobserved holding actions. Closed-loop counterfactual policy claims therefore require a validated simulator, benchmark simulator, intervention logs, natural experiments, or field trials; this is why the RESCO traffic-signal benchmark is reported as a counterfactual stress test rather than as bus-field evidence.

## 4 Method

### 4.1 Dense H2O<sup>+</sup>-style residual baseline

The H2O<sup>+</sup>-style baseline fits one dense residual model over all transition outputs:

$$\hat{\Delta}_{\text{dense}}(s, a) = \phi_{\text{dense}}(s, a)^\top B,$$

where  $\phi_{\text{dense}}$  includes normalized state/action features, squares, action interactions, and selected pairwise products. At evaluation time the corrected prediction is

$$\hat{y}_{\text{H2O}^+}(s, a) = f_0(s, a) + \hat{\Delta}_{\text{dense}}(s, a).$$

The baseline is intentionally strong: it can use many cross terms and is fit on all source-city transitions. It is nevertheless an H2O<sup>+</sup>-style dense residual transfer surrogate rather than a full reproduction of the original H2O+ policy-learning stack; Table 20 documents the implementation differences.

### 4.2 Causal-factored residual transfer

CFCMT decomposes the residual by output mechanism:

$$\hat{\Delta}_j(s, a) = \phi_j(s, a)^\top \beta_j, \quad j \in \{1, \dots, 9\}.$$

The feature map  $\phi_j$  is chosen according to the output mechanism. Headway and gap residuals use current headways, target headway, speed, action, and time. Waiting and dwell residuals use waiting passengers, target headway, speed, station fraction, action, and time. Speed residuals use current speed, event time, and station position. Reward residuals use policy-relevant headway deviations, waiting passengers, gap, speed, action, and time. The corrected prediction is

$$\hat{y}_{\text{CFCMT}}(s, a) = f_0(s, a) + \left( \hat{\Delta}_1(s, a), \dots, \hat{\Delta}_9(s, a) \right).$$

All reported models use ridge-regularized sufficient statistics, so the experiment can be reproduced without stochastic neural training.

### 4.3 Theoretical motivation

The theory we need is not a proof of causal identification. Instead, it explains why a mechanism-factored residual and a guarded source ensemble are natural for cross-city transfer when target transition labels are unavailable. Let  $x = (s, a)$  and let

$$\delta_c(x) = f_c(x) - f_0(x)$$

denote the simulator residual for city  $c$ . For a residual predictor  $g$ , define the target risk

$$R_t(g) = \mathbb{E}_{x \sim P_t} [\|\delta_t(x) - g(x)\|_2^2],$$

where  $P_t$  is the target-city decision-state distribution induced by the offline validation set. Partition the nine outputs into mechanism groups  $\mathcal{M}_1, \dots, \mathcal{M}_K$ , such as headway/gap, demand/dwell/reward, and speed. For  $g = (g_1, \dots, g_K)$  and  $\delta_c = (\delta_{c,1}, \dots, \delta_{c,K})$ , squared-error risk decomposes exactly:

$$R_t(g) = \sum_{k=1}^K \mathbb{E}_{x \sim P_t} [\|\delta_{t,k}(x) - g_k(x)\|_2^2].$$

Thus each mechanism contributes an auditable term to total error. This does not imply that the groups are causally identified, but it gives a formal reason to fit, validate, and report mechanism-wise residual errors rather than only a dense total error.

**Parent-set invariance assumption.** The operative assumption is weaker than full causal identification. For each output mechanism  $k$ , there exists a small parent set  $Pa_k(x)$  such that the residual map  $\delta_{c,k}$  varies across cities mainly through mechanism-specific coefficients, context bias, or source weights, while irrelevant city-specific nuisance features do not improve target risk after source validation. The sparse parent sets in CFCMT encode this assumption for headway/gap, demand/dwell/reward, and speed outputs. The assumption can fail: if the target generator or real operation shifts an omitted parent, the factored model can underfit. This is why the experiments include wrong-grouping, action/time-only, dense matched-feature, source-subset, and generator-bias diagnostics, and why the paper avoids claiming causal identification from observational static data alone.

**Mechanism-shift bound.** Let  $\bar{\delta}_{w,k} = \sum_{c \in \mathcal{C}_{\text{src}}} w_c \delta_{c,k}$  be the weighted source residual for mechanism  $k$ . Suppose the fitted mechanism predictor satisfies

$$\|\hat{g}_k - \bar{\delta}_{w,k}\|_{L_2(P_t)} \leq \epsilon_k, \quad \|\bar{\delta}_{w,k} - \delta_{t,k}\|_{L_2(P_t)} \leq \Delta_k(w),$$

where  $\epsilon_k$  is estimation/approximation error and  $\Delta_k(w)$  is the cross-city mechanism shift left after source weighting. By the triangle inequality,

$$\sqrt{R_t(\hat{g})} \leq \left( \sum_{k=1}^K [\epsilon_k + \Delta_k(w)]^2 \right)^{1/2}.$$

The bound clarifies the intended advantage: if only some mechanisms shift strongly across cities, a factored model can localize transfer error to those mechanisms. A dense residual can still perform well, but its error is harder to attribute and may mix source-specific headway, demand, dwell, and speed shifts in a single parameter block.

**Source-ensemble risk bound.** For source-city experts  $g_c$ , the ensemble predictor  $g_w = \sum_c w_c g_c$  has a convexity bound under squared loss:

$$R_t(g_w) = \mathbb{E}_{P_t} \left[ \left\| \sum_c w_c (\delta_t - g_c) \right\|_2^2 \right] \leq \sum_c w_c R_t(g_c), \quad w_c \geq 0, \quad \sum_c w_c = 1.$$

The unavailable quantity is  $R_t(g_c)$ . The target-label-free source-weighting rule therefore uses unlabeled target summaries as a proxy for which source-city expert is likely to have low target risk. The dominance guard  $\max_c w_c(t) \geq 0.5$  is a conservative negative-transfer control: an ensemble is allowed to override the rigid factored residual only when the unlabeled target summary identifies at least one clearly close source; otherwise the method falls back to the rigid source-pooled model.

**Source-only selector bound.** Let  $q$  index a candidate target-summary feature family, and let  $S(q)$  be its leave-one-source validation score. Let  $R_t(q)$  denote the target risk of the full guarded procedure using feature family  $q$ . If the source-fold score approximates the target ranking up to

$$|R_t(q) - S(q)| \leq \gamma + \varepsilon \quad \text{for all candidate families } q,$$

where  $\gamma$  is the source-to-target representativeness gap and  $\varepsilon$  is finite-sample validation error, then the selected family

$$\hat{q} = \arg \min_q S(q)$$

satisfies

$$R_t(\hat{q}) \leq \min_q R_t(q) + 2(\gamma + \varepsilon).$$

The condition is not guaranteed by the data; it is the assumption being tested by source-subset and feature-family diagnostics. It is nevertheless important methodologically: feature-family selection can be done without target transition labels, and any failure mode is visible as a source-representativeness problem rather than target-label tuning.

#### 4.4 Unlabeled target source selection

For each city we compute a static descriptor  $z_c$  from speed quantiles, headway quantiles, route-distance quantiles, peak-demand hour, and the 24-hour demand-share vector. For a target city  $t$  and source city  $c$ , the distance is computed after feature-wise standardization over the source-plus-target set:

$$d(c, t) = \frac{1}{\sqrt{m}} \left\| \frac{z_c - \mu}{\sigma} - \frac{z_t - \mu}{\sigma} \right\|_2.$$

The source weight is

$$w_c = \frac{\exp(-d(c, t)/T) + \lambda \max_{c'} \exp(-d(c', t)/T)}{\sum_{k \in \mathcal{C}_{\text{src}}} [\exp(-d(k, t)/T) + \lambda \max_{c'} \exp(-d(c', t)/T)]}.$$

The default values are  $T = 1.0$  and  $\lambda = 0.05$ . The source sufficient statistics are rescaled so that these weights control the effective number of source transitions rather than simply reflecting raw city size.

#### 4.5 Rigid-first source-city ensemble

The full-module variant starts from the causal-factored ridge prediction rather than from the uncorrected simulator:

$$\hat{\mathbf{y}}_{\text{rigid}}(s, a) = f_0(s, a) + \hat{\Delta}_{\text{CFCMT}}(s, a).$$

It then fits a second residual on the remaining source error,

$$e_c(s, a) = \mathbf{y}_c^{\text{target}} - \hat{\mathbf{y}}_{\text{rigid}}(s, a),$$



using dense features, deterministic local-graph features, or their concatenation. A candidate correction has the form

$$\hat{\mathbf{y}}_k(s, a) = \hat{\mathbf{y}}_{\text{rigid}}(s, a) + \alpha \eta_k(s, a), \quad \alpha \in \{0.25, 0.50, 0.75, 1.00\}.$$

For the source-city ensemble candidate, the second residual is fit separately on each source city and mixed by target similarity:

$$\eta_k^{\text{ens}}(s, a; t) = \sum_{c \in \mathcal{C}_{\text{src}}} w_c(t) \eta_{k,c}(s, a).$$

The source-summary feature set is not chosen by target performance. For each candidate summary family (simulator-output only, observation only, observation plus simulator output, local-only, and full summary), the same leave-one-source folds below are used to score the selected residual candidate. The summary family with the best source-fold score is then used to compute  $w_c(t)$  for the held-out target. None of these variants uses target transition labels.

**Source-summary feature-set selector.** The guarded ensemble uses the following target-label-free selection procedure.

1. Define the candidate summary families listed in Table 21.
2. For each candidate family, run the same leave-one-source folds used for residual-candidate selection.
3. Within each fold, select the residual candidate by source-fold total-MSE ratio to rigid CFCMT, subject to the ensemble dominance guard.
4. Score the summary family by the selected candidate’s mean source-fold ratio; lower is better.
5. Break ties by preferring simulator-output summaries, then observation-plus-simulator, observation-only, full, local-only, and uniform weighting.
6. Freeze the selected summary family before target evaluation and compute target source weights from unlabeled target summaries only.

Candidate selection uses leave-one-source validation. For each held-out source city  $h$ , we train on  $\mathcal{C}_{\text{src}} \setminus \{h\}$ , evaluate on  $h$ , and record each candidate’s total-MSE ratio to rigid CFCMT. The selected candidate must have stable source-fold gains. Ensemble candidates are additionally guarded: they may override the conservative rigid/global candidate only when the target similarity weights have a dominant source,  $\max_c w_c(t) \geq 0.5$ . This guard prevents a diffuse source mixture from overriding rigid CFCMT when the unlabeled target summary does not identify a close source city.

## 4.6 Traffic-signal phase-transfer stress test

For the RESCO stress test, the same transfer principle is applied to phase-local traffic-signal models. For each traffic light and candidate green phase, a generic simulator prior predicts the next local queue cost. The dense baseline fits one pooled residual over all phase-local features, while the CFCMT variant decomposes the residual into queue-propagation, downstream-spillback, and phase-pressure feature groups. At each decision time, model-based policies evaluate the feasible SUMO-defined green phases and select

$$a_t^* = \arg \min_{a \in \mathcal{A}(i)} \hat{c}(x_{i,t}, a),$$



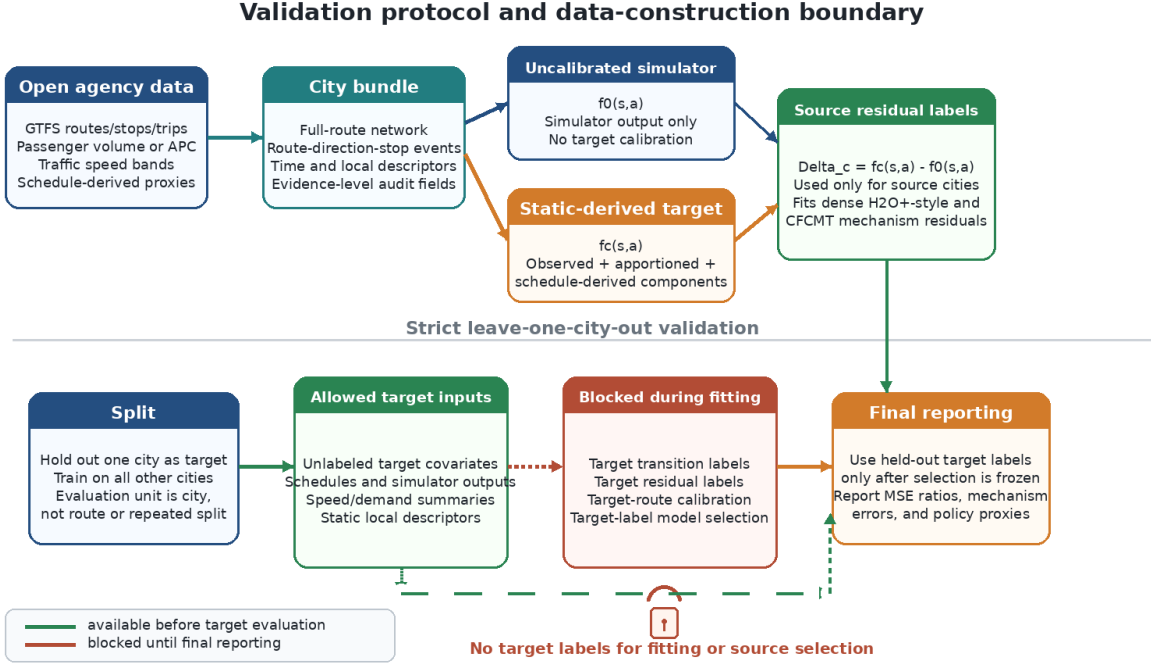


Figure 2: Validation protocol and data-construction boundary. Open agency data are converted into full-route city bundles, paired simulator outputs  $f_0(s, a)$  and static-derived targets  $f_c(s, a)$ , and source-city residual labels. In strict leave-one-city-out validation, target covariates and unlabeled summaries may be used for source weighting and feature-set selection, but held-out target transition labels are used only after selection is frozen for final reporting.

is total MSE averaged over the nine transition outputs. Ratios below 1 indicate improvement over the dense H2O<sup>+</sup>-style residual baseline.

**Single-source zero-shot diagnostic.** We also run every ordered one-source-to-one-target city pair. Only the source city provides residual labels; the target city provides the same network and static summary inputs used by the target-label-free selector, but no target transition labels before reporting. This diagnostic is intentionally harder and is not expected to be uniformly positive. Its role is to establish the minimum-information baseline before adding source diversity or target passive history.

**Target offline adaptation / few-shot curve.** The target-route budget sweep is interpreted as offline adaptation, not zero-shot transfer. For each held-out target city, 0%, 1%, 5%, 10%, 25%, or 100% of target route groups are added to the source sufficient statistics, and evaluation is performed on the remaining target route groups except at the 100% in-sample oracle reference. This approximates the practical deployment setting in which a new city has a small amount of passive AVL/APC or schedule-derived historical labels before control deployment. We additionally run same-information few-shot adaptation baselines motivated by AdaRL and FANsRL. The AdaRL-style baseline uses the target budget to select which mechanism groups should be adapted and then refits only those groups. The FANsRL-style baseline keeps the transferred mechanism parameters fixed and estimates a low-dimensional per-output target context bias. These are not direct execu-

| city              | demand.evidence               | traffic.evidence     | lines | transitions | uncal_mse | line_mse.p25 | line_mse.p50 | line_mse.p75 |
|-------------------|-------------------------------|----------------------|-------|-------------|-----------|--------------|--------------|--------------|
| Singapore         | observed stop OD/PV monthly   | observed speed bands | 789   | 1243488     | 2256.5339 | 220.173      | 324.7508     | 541.4336     |
| Austin / CapMetro | schedule proxy                | schedule derived     | 234   | 310272      | 1301.3027 | 242.2482     | 615.8957     | 3043.8061    |
| Halifax Transit   | route APC apportioned         | schedule derived     | 176   | 266448      | 1111.9223 | 199.2164     | 620.2908     | 1614.3016    |
| MBTA              | stop board/alight apportioned | schedule derived     | 940   | 969360      | 1216.3716 | 180.9625     | 459.6093     | 1040.0624    |

Table 2: Target-construction audit. The evidence columns identify observed, apportioned, and schedule-derived components; route-level residual quantiles show that simulator-target gaps are heterogeneous rather than a single constant correction.

tions of the original CartPole/Atari or MuJoCo codebases; they are bus-residual evaluators that use the same target labels and the same source-city statistics as CFCMT.

**One-step policy proxy.** The one-step policy experiment uses the same source-target splits and evaluates action selection under headway-stress states. We report reward/regret and traffic-domain metrics. Bunching rate is the fraction of decisions in which either adjacent headway is below 0.5 times the target headway. Large-gap rate is the fraction in which either adjacent headway exceeds 1.5 times the target headway.

**Reviewer-facing robustness checks.** We include diagnostics targeted at common review objections. First, a target-construction audit separates observed, apportioned, schedule-proxy, and deterministic target components. Second, the target-route offline-adaptation sweep adds 0%, 1%, 5%, 10%, 25%, or 100% of held-out target-city routes to the source sufficient statistics and evaluates on the remaining target routes; the 100% point is an in-sample oracle reference. Third, per-mechanism errors decompose total MSE into headway/gap, demand/dwell/reward, and speed groups. Fourth, mechanism ablations compare the proposed parent sets against wrong mechanism groupings, action/time-only features, and dense matched-capacity variants. Fifth, source-weight sensitivity varies  $T \in \{0.5, 1.0, 2.0\}$  and  $\lambda \in \{0, 0.05, 0.1\}$ , while source-subset and pairwise-source checks remove major source families. Sixth, generator robustness perturbs output groups or adds measurement-noise terms to the evaluation sufficient statistics.

**Full-module guarded-ensemble validation.** The main paper-suite tables use deterministic sufficient-statistics variants for reproducibility. We additionally report a full-module diagnostic that adds rigid-first residual deltas, deterministic static-local features, source-city leave-one validation, and the guarded source-city ensemble. For the ensemble weights, a source-only feature-set selector chooses among simulator-output-only, observation-only, observation-plus-simulator, local-only, full-summary, and uniform-weight alternatives before target evaluation. We also add a same-information dense H2O<sup>+</sup> baseline that uses the same unlabeled target summary to weight source-city dense residual experts. This diagnostic uses six configured cross-city splits: the four strict leave-one-city-out targets plus two source-family transfer splits used to stress the source selector. Because this diagnostic contains repeated target cities, it is interpreted as a selector stress test rather than as six independent city-level targets.

**External real APC/AVL validation.** To partially address the static-derived target limitation, we add an external Austin/CapMetro validation slice from public raw APC stop-event data. The July–December 2021 and August 2016 files contain observed stop events with boardings, alightings, vehicle load, dwell time, vehicle position, route, trip, stop, and vehicle identifiers. We filter records with valid stop, trip, GPS, and import-quality flags, reconstruct stop-level headways by route-direction-stop event times, and evaluate passive action-0 one-step dynamics against the observed

next stop event. The no-correction baseline in this experiment is a state-informed passive transition using the APC-derived current state, not a schedule-only free-running simulator. This is real historical APC/AVL validation, but it is still offline passive validation rather than a field control deployment because counterfactual hold actions are not observed. In observational AVL/APC logs, a counterfactual hold outcome would require knowing what would have happened in the same state under a hold action that was not actually taken; such labels require randomized or quasi-randomized field interventions, an experimentally validated simulator, or another credible causal design. We also run a few-shot APC adaptation sweep at 0%, 1%, 5%, 10%, and 25% of Austin line groups. This sweep compares source-only transfer, source-plus-target residual refitting, target-only residual fitting, a residual gate that learns how much of the source correction to trust, and a target residual-bias adapter. The gate and bias-adapter variants are lightweight AdaRL/FANsRL-style checks of what changes should be adapted; they are not full reimplementations of those neural RL systems. Because route-random budgets are not the most natural deployment unit, we also add a chronological target-history sweep using the first 0, 1, 6, 24, or 168 hours of Austin APC history for adaptation and all later APC transitions for evaluation.

**Auxiliary executable rollout.** Sampled BusSimEnv rollout is reported as an auxiliary sanity check on runnable line-level environments generated from each city bundle. For this rollout, each policy is evaluated on the same line-level random seed, including Python signal-delay randomness, and live observations are mapped back to the static-transition feature schema before model scoring. The policy score uses the BusSimEnv linear headway reward rather than the static one-step reward.

**Full-line libsumo holding diagnostic.** We also generate executable SUMO scenarios with explicit bus-stop dwell points for every validated line in each city bundle. The diagnostic uses a four-hour window with at most eight departures per line and evaluates no-hold, dense H2O<sup>+</sup>-MPC, weighted CFCMT-MPC, threshold equalization, retuned Daganzo, and a target-summary gated Daganzo prior. The target-summary prior uses only static target route and schedule summaries to choose whether a conservative or aggressive Daganzo gain is appropriate; it does not use target transition labels or the held-out closed-loop result for fitting. This rollout is a counterfactual simulator diagnostic, not field-calibrated evidence, but it tests whether the proposed mechanism-state idea should control a safe prior or a free-form residual action.

**Native-protocol RL cross-checks.** Finally, we run official RESCO and patched LibSignal RL baselines under each library’s native state, reward, action, and logging definitions. The official RESCO table trains IDQN and upstream-supported MPLight profiles for 100 episodes and evaluates five test episodes on the extended RESCO scenario set. The LibSignal table trains DQN, PressLight, FRAP, and MPLight for 100 episodes on sumo1x1, sumo1x3, and sumo4x4, with a final 3600-step test. These rows answer a reviewer-facing reproducibility question: standard RL runners are connected and run at full budget. They are not same-protocol comparisons against CFCMT, because their observation spaces, rewards, action timing, and reported metrics differ from the phase-MPC stress-test protocol.

## 6 Results

### 6.1 Strict leave-one-city-out dynamics

Figure 3 and table 4 show the primary dynamics result. Unweighted CFCMT beats the dense H2O<sup>+</sup>-style residual baseline on all four held-out cities, with mean total-MSE ratio 0.759. Similarity-

| experiment                   | paper_role                           | unit                   | scope_note                                                                   |
|------------------------------|--------------------------------------|------------------------|------------------------------------------------------------------------------|
| strict_leave_one_city_out    | primary cross-city dynamics evidence | city target            | four unique targets; no duplicated target counted as independent             |
| generator_robustness         | bias/noise defense                   | scenario x city target | static-derived target perturbation; not real AVL/APC trajectory ground truth |
| source_subset_robustness     | data-source confound defense         | city subset            | reports excluding Singapore and excluding Austin proxy-demand subsets        |
| cross_city_policy_validation | primary one-step policy evidence     | city target            | one-step lookahead, not live SUMO rollout                                    |
| sampled_rollout              | auxiliary sanity check only          | BusSimEnv episode      | 50 policy episodes across runnable line envs; do not use as primary claim    |

Table 3: Experimental scope and paper role. The primary evidence is strict city-level leave-one-out dynamics; one-step policy and sampled rollout results are secondary control-facing diagnostics.

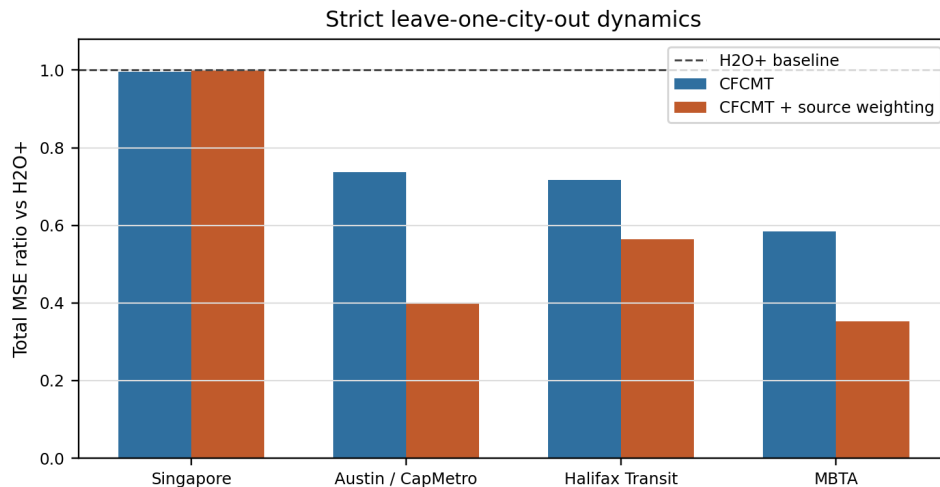


Figure 3: Strict leave-one-city-out dynamics validation. Bars show total-MSE ratio relative to  $H2O^+$ . The dashed line is parity with  $H2O^+$ .

weighted CFCMT also wins on all four targets and improves the mean ratio to 0.578. The largest gains occur for Austin, Halifax, and MBTA. Singapore is the hardest target: unweighted CFCMT is only slightly better than  $H2O^+$ , and similarity weighting is slightly worse than unweighted CFCMT while still remaining below the  $H2O^+$  baseline.

## 6.2 Target-information ladder

Table 5 separates the regimes that are easy to confuse in cross-city transit transfer. The single-source zero-shot diagnostic is deliberately brittle: across the 12 ordered source-target pairs, CFCMT wins against  $H2O^+$  on 6 pairs with mean ratio 0.927. This shows that “train on one city, deploy in another” is possible in some directions but not a stable main claim. Adding source diversity changes the result: the multi-source target-label-free setting wins on all four held-out cities with mean weighted ratio 0.578. With small target-route historical labels, the same framework becomes target offline adaptation rather than zero-shot transfer; the 1%–25% route-budget points remain below  $H2O^+$ -style dense adaptation on three of four strict targets. The 100% target-route point is kept only as an upper-bound reference.

| regime                         | source_cities     | target_labels       | target_side_info         | splits | wins | mean_ratio_vs_h2oplus | paper_role                     |
|--------------------------------|-------------------|---------------------|--------------------------|--------|------|-----------------------|--------------------------------|
| single-source zero-shot        | 1                 | 0                   | network/static summaries | 12     | 6    | 0.9272                | minimum-information diagnostic |
| multi-source target-label-free | all except target | 0                   | network/static summaries | 4      | 4    | 0.5782                | primary cross-city result      |
| target offline adaptation 0.01 | all except target | 0.01 route fraction | target passive labels    | 4      | 3    | 0.5801                | few-shot deployment curve      |
| target offline adaptation 0.05 | all except target | 0.05 route fraction | target passive labels    | 4      | 3    | 0.5831                | few-shot deployment curve      |
| target offline adaptation 0.1  | all except target | 0.1 route fraction  | target passive labels    | 4      | 3    | 0.6094                | few-shot deployment curve      |
| target offline adaptation 0.25 | all except target | 0.25 route fraction | target passive labels    | 4      | 3    | 0.6596                | few-shot deployment curve      |
| target-route oracle            | all except target | 1 route fraction    | all target route labels  | 4      | 3    | 0.7242                | upper-bound reference          |

| target_city       | source_cities                                                     | target_transitions | h2oplus_total_mse | cfcmnt_total_mse | weighted_cfcmnt_total_mse | cfcmnt_vs_h2oplus_ratio | weighted_cfcmnt_vs_h2oplus_ratio | weighted_cfcmnt_vs_unweighted_ratio |
|-------------------|-------------------------------------------------------------------|--------------------|-------------------|------------------|---------------------------|-------------------------|----------------------------------|-------------------------------------|
| Singapore         | austin, capmetro, all, halifax, transit, all, mbta, all           | 1243488            | 2254.8548         | 2246.6693        | 2251.7081                 | 0.9964                  | 0.9986                           | 1.0022                              |
| Austin / CapMetro | singapore, ita, all, halifax, transit, all, mbta, all             | 310272             | 91.9724           | 67.8254          | 36.6531                   | 0.7375                  | 0.3985                           | 0.5404                              |
| Halifax Transit   | singapore, ita, all, austin, capmetro, all, mbta, all             | 266448             | 97.1175           | 69.5992          | 54.7536                   | 0.7166                  | 0.5638                           | 0.7867                              |
| MBTA              | singapore, ita, all, austin, capmetro, all, halifax, transit, all | 969660             | 318.9925          | 186.4177         | 112.2698                  | 0.5844                  | 0.352                            | 0.6022                              |

Table 4: Strict leave-one-city-out dynamics results. Ratios below 1 indicate lower total MSE than H2O<sup>+</sup>.

| split                                             | target_city     | selected_candidate                                 | selected_is_ensemble | max_source_weight | h2oplus_mse | rigid_cfcmnt_mse | guarded_mse | guarded_ratio_vs_h2oplus | guarded_ratio_vs_rigid |
|---------------------------------------------------|-----------------|----------------------------------------------------|----------------------|-------------------|-------------|------------------|-------------|--------------------------|------------------------|
| source, open, na, to, singapore                   | singapore ita   | dense, static, local, delta, alpha, dp75           | False                | 0.465             | 2376.1427   | 2388.7628        | 2367.5165   | 0.9964                   | 0.9911                 |
| source, singapore, austin, halifax, to, mbta      | mbta            | ensemble, dense, static, local, delta, alpha, dp50 | True                 | 0.5909            | 209.6939    | 116.4969         | 93.8321     | 0.4475                   | 0.8054                 |
| leave, one, city, out, all: singapore, ita, all   | singapore ita   | dense, static, local, delta, alpha, dp75           | False                | 0.465             | 2376.1427   | 2388.7628        | 2367.5165   | 0.9964                   | 0.9911                 |
| leave, one, city, out, all: austin, capmetro, all | austin capmetro | ensemble, dense, static, local, delta, alpha, dp25 | True                 | 0.5905            | 63.8731     | 47.6396          | 40.7936     | 0.6387                   | 0.8563                 |
| leave, one, city, out, all: halifax, transit, all | halifax transit | rigid, cfcmnt                                      | False                | 0.3749            | 73.5689     | 53.1982          | 53.1982     | 0.7231                   | 1.0                    |
| leave, one, city, out, all: mbta, all             | mbta            | ensemble, dense, static, local, delta, alpha, dp50 | True                 | 0.5909            | 209.6939    | 116.4969         | 93.8321     | 0.4475                   | 0.8054                 |

Table 6: Full-module guarded source-ensemble diagnostic. This table uses six configured selector-stress splits, including repeated target cities; it is therefore a source-selection diagnostic rather than the four-target strict leave-one-city-out count in Table 4.

Table 5: Target-information ladder. Zero-shot rows use no target transition labels before reporting; target offline adaptation rows use the stated fraction of passive target route labels and evaluate on held-out target routes except for the oracle row.

### 6.3 Guarded full-module source ensemble

The full-module diagnostic tests whether the rigid CFCMT residual can be improved without target-label calibration. The best selected variant is a rigid-first source-gated delta with a guarded source-city ensemble. It uses unlabeled target-city simulator summaries to weight source-city residual experts and to decide whether an ensemble is allowed; target transition labels remain held out. The source-summary feature set is selected only by source-fold diagnostics, not by target performance; the selected feature sets are reported in Table 27. Across the six configured selector-stress splits, the guarded selector lowers mean total MSE from 884.853 for the dense H2O<sup>+</sup>-style baseline and 851.893 for rigid CFCMT to 836.115, with 6/6 wins against H2O<sup>+</sup>. A same-information dense H2O<sup>+</sup> source ensemble, using the same unlabeled target summaries for source weighting, improves to 846.466 but remains weaker than the guarded mechanism-factored ensemble. Because our dense baseline is a reproducible H2O<sup>+</sup>-style residual-transfer surrogate rather than a full H2O<sup>+</sup> reproduction, Table 20 lists the implemented and omitted components.

The guard is important. MBTA and Austin have a dominant similar source under the selected target-summary descriptor, so the ensemble is allowed and improves the selected MSE substantially. Halifax has diffuse source weights, so the guard falls back to rigid CFCMT and avoids the unguarded ensemble regression. Per-mechanism hard splicing is weaker in this diagnostic, with mean total MSE 864.592, because mechanism-level source-fold improvements do not reliably transfer to every target. Additional same-information baselines, selector ablations, threshold sensitivity, target-label leakage checks, feature-set ablations, and oracle-gap diagnostics are reported in Tables 22, 23, 25, 26, 29 and 30.

### 6.4 One-step policy proxy

The strict one-step policy summary is more mixed but still favorable. CFCMT improves reward/regret on three of four targets, improves mean absolute headway error on three of four, and lowers bunching rate on all four targets. The mean regret ratio is 0.876, and the mean bunching-rate ratio is 0.644. Halifax is the main policy exception: CFCMT lowers headway error slightly but has worse reward/regret than H2O<sup>+</sup>. This is consistent with a limited control-facing claim:

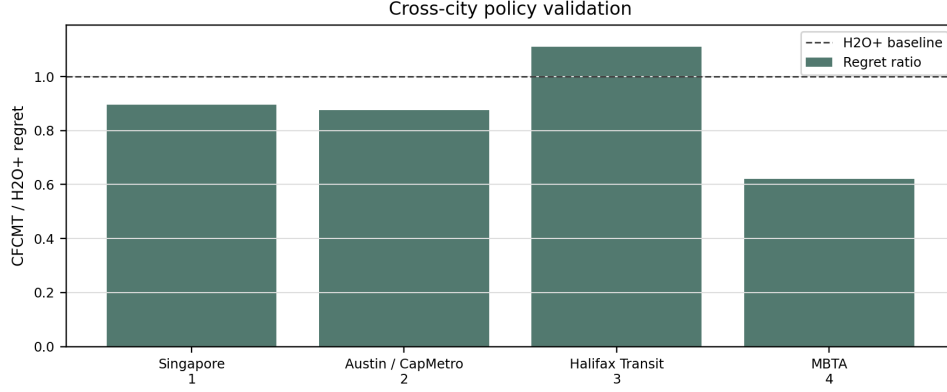


Figure 4: Strict leave-one-city-out one-step policy regret ratio. Ratios below 1 favor CFCMT over H2O<sup>+</sup>.

| split              | target_city       | h2o_plus_reward | cfcmt_reward | cfcmt_reward_gain | cfcmt_regret_ratio | cfcmt_headway_error_ratio | h2o_plus_bunching_rate | cfcmt_bunching_rate | cfcmt_bunching_rate_ratio | h2o_plus_large_gap_rate | cfcmt_large_gap_rate | cfcmt_large_gap_rate_ratio | cfcmt_hold_time_delta |
|--------------------|-------------------|-----------------|--------------|-------------------|--------------------|---------------------------|------------------------|---------------------|---------------------------|-------------------------|----------------------|----------------------------|-----------------------|
| leave-one-city-out | Singapore         | -0.3338         | -0.3337      | 0.0001            | 0.9999             | 1.0007                    | 0.0024                 | 0.0024              | 0.9991                    | 0.0036                  | 0.0036               | 1.0                        | -0.3597               |
| leave-one-city-out | Austin / CapMetro | -0.3326         | -0.332       | 0.0007            | 0.9993             | 0.9995                    | 0.0                    | 0.0                 | 0.9995                    | 0.0001                  | 0.0                  | 0.0                        | -0.3609               |
| leave-one-city-out | Halifax Transit   | -0.3307         | -0.3312      | -0.0005           | 1.1113             | 0.9998                    | 0.0                    | 0.0                 | 0.9355                    | 0.0                     | 0.0                  | 1.0                        | 1.4814                |
| leave-one-city-out | MBTA              | -0.3429         | -0.338       | 0.0049            | 0.6201             | 0.9967                    | 0.0004                 | 0.0004              | 0.6396                    | 0.0002                  | 0.0002               | 0.3174                     | -1.1821               |

Table 7: Strict leave-one-city-out one-step policy-proxy metrics. The table includes reward, regret ratio, headway error ratio, bunching, large-gap rate, and hold-time difference.

mechanism-factored transfer improves cross-city dynamics strongly and often improves one-step policy-proxy quality, but it is not a closed-loop operational dominance result.

## 6.5 Source weighting is stable under a small grid

The default source weighting uses  $T = 1.0$  and  $\lambda = 0.05$ . Across the  $3 \times 3$  grid, all nine settings beat H2O<sup>+</sup> on all six configured cross-city splits. The mean weighted-CFCMT/H2O<sup>+</sup> total-MSE ratio ranges only from 0.6025 to 0.6149. The best setting in the grid is  $T = 0.5, \lambda = 0$ , but the default is not selected by post-hoc optimization and remains within the same narrow band.

## 6.6 Target offline adaptation and mechanism-level errors

The target-route offline-adaptation sweep tests whether a dense residual baseline with the same target passive-label budget quickly erases the no-target-label CFCMT advantage. At 0% target labels, weighted CFCMT has mean ratio 0.578 against the dense source-only baseline. When 1%, 5%, 10%, and 25% of target routes are added to the dense baseline, the uncalibrated weighted-CFCMT/H2O ratio remains below 1 on three of four strict targets, with mean ratios 0.581, 0.586, 0.617, and 0.677, respectively. When CFCMT is allowed to use the same target route budget through the mechanism-weighted sufficient statistics, the corresponding mean ratios are 0.580, 0.583, 0.609, and 0.660. The 100% target-route point is an in-sample oracle reference and should not be interpreted as deployable no-calibration evidence.

The same-information adaptation-baseline comparison gives a sharper design lesson. AdaRL-style selected refit is nearly identical to full CFCMT refit because the target calibration slice usually selects all mechanism groups. By contrast, the low-dimensional FANsRL-style context-bias adapter is better than full refit on average at every positive non-oracle budget, with mean ratios 0.559, 0.545, 0.572, and 0.629 against the dense adapted baseline. It wins on four targets at the 1% budget and on three of four targets at larger route budgets. Thus the best few-shot variant is not unconstrained



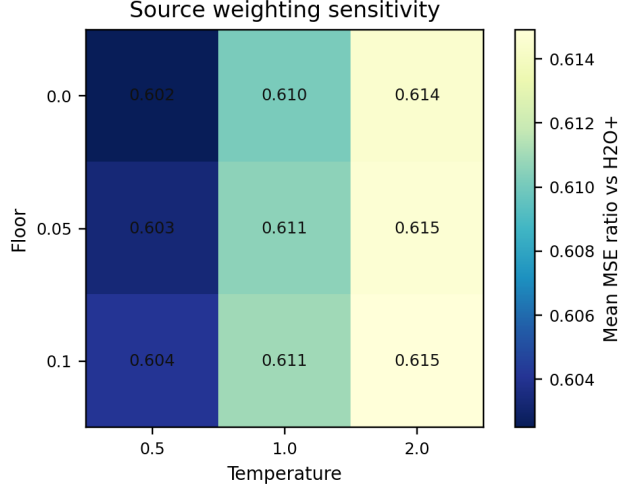


Figure 5: Source-weighting sensitivity. Each cell reports mean total-MSE ratio versus  $H2O^+$  across configured cross-city splits.

| budget | N | cal.lines | cal.trans | cfcmt_refit | adarl_refit | adarl_bias | fansrl_bias | fansrl_wins | refit_wins | oracle |
|--------|---|-----------|-----------|-------------|-------------|------------|-------------|-------------|------------|--------|
| 0.0    | 4 | 0.0       | 0.0       | 0.5782      | 0.5782      | 0.5782     | 0.5782      | 4           | 4          | False  |
| 0.01   | 4 | 3.5       | 3972.0    | 0.5801      | 0.5801      | 0.5585     | 0.5585      | 4           | 3          | False  |
| 0.05   | 4 | 23.0      | 30468.0   | 0.5831      | 0.5832      | 0.5472     | 0.5446      | 3           | 3          | False  |
| 0.1    | 4 | 50.5      | 61704.0   | 0.6094      | 0.6094      | 0.5718     | 0.5717      | 3           | 3          | False  |
| 0.25   | 4 | 132.75    | 167712.0  | 0.6596      | 0.6616      | 0.6318     | 0.6289      | 3           | 3          | False  |
| 1.0    | 4 | 534.75    | 697392.0  | 0.7242      | 0.7244      | 0.7222     | 0.7221      | 3           | 3          | True   |

Table 8: Same-information few-shot adaptation baselines. Ratios are total MSE relative to the dense  $H2O^+$ -style baseline using the same target route-label budget. The 1.0 budget is an in-sample oracle reference.

target refitting; it is mechanism-factored transfer plus a low-dimensional target context correction.

Per-mechanism errors show that the total-MSE advantage is not driven by one output group only. Weighted CFCMT improves all three mechanism groups on average: headway/gap ratio 0.574, demand/dwell/reward ratio 0.771, and speed ratio 0.668. Headway/gap is the largest contributor to the main gain, which is consistent with the holding-control objective.

## 6.7 Robustness and boundary conditions

Generator robustness supports the main trend but also identifies failure modes. Weighted CFCMT beats  $H2O^+$  on every city target in five of seven perturbation/noise scenarios. Under headway-gap bias and mixed-mechanism bias, the result weakens to three of four targets. Thus the evidence does not say that CFCMT is invariant to any target-generation change; it says that the result is robust to several plausible perturbations.

Mechanism ablations support the mechanism-specific interpretation, with a caveat. Wrong mechanism groupings and action/time-only features are much worse than  $H2O^+$ , with mean ratios above 7. Dense matched sparse features are close to full CFCMT, with mean ratio 0.772, which means the result should be described as evidence for mechanism-wise sparse transfer rather than definitive causal identification.

The source-subset check is the strongest limitation. With all four cities, weighted CFCMT has

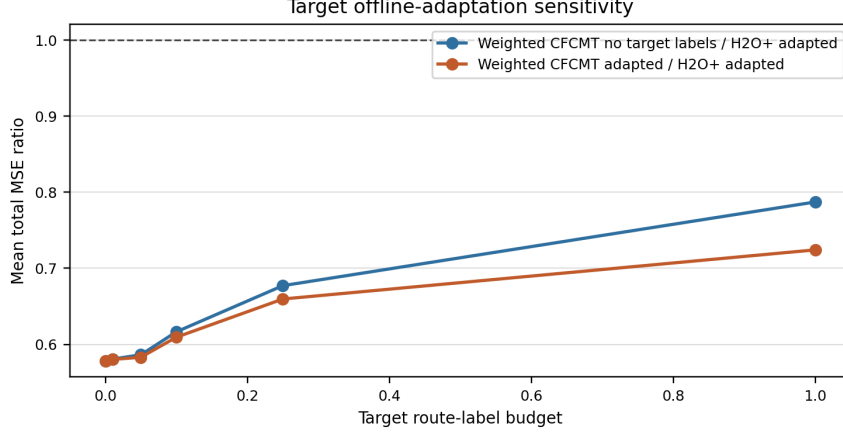


Figure 6: Target-route offline-adaptation budget sensitivity. Ratios compare weighted CFCMT with no target labels or the same target route-label budget against a dense H2O<sup>+</sup>-style baseline using that target budget.

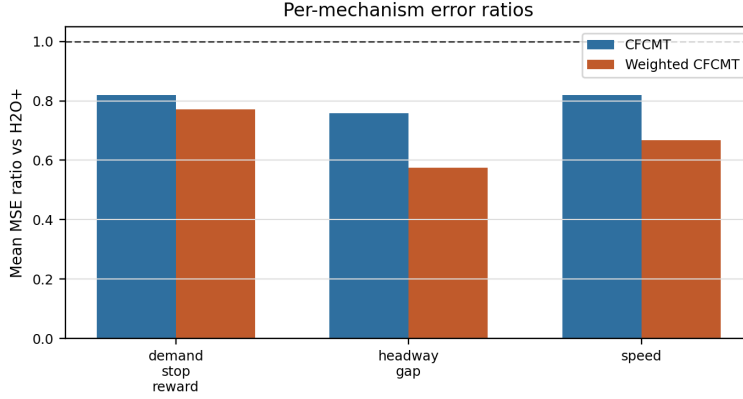


Figure 7: Per-mechanism mean MSE ratios under strict leave-one-city-out dynamics validation.

mean ratio 0.578. Excluding Austin remains favorable with mean ratio 0.663. Excluding Singapore makes the result nearly tied with H2O<sup>+</sup>, with mean ratio 0.998. The two-city Halifax-MBTA pair is slightly worse than H2O<sup>+</sup>, with ratio 1.042. This indicates that source diversity matters: the method should be framed as multi-source cross-city transfer, not as guaranteed transfer between every pair of cities.

Route-level bootstrap confidence intervals reinforce the same boundary. Unweighted CFCMT has a route-bootstrap ratio confidence interval below 1 for all four targets. Weighted CFCMT has intervals below 1 for Austin, Halifax, and MBTA, but Singapore crosses parity with interval [0.993, 1.002]. The conservative reading is that source weighting improves the mean strongly, while Singapore remains the least secure target.

## 6.8 External real APC/AVL validation

The public Austin raw APC validation is the strongest non-static check in the current paper. Across two real APC/AVL datasets, the evaluation uses 11.57 million observed stop-event transitions after quality filtering. Weighted CFCMT beats the dense H2O<sup>+</sup>-style residual transfer baseline in both

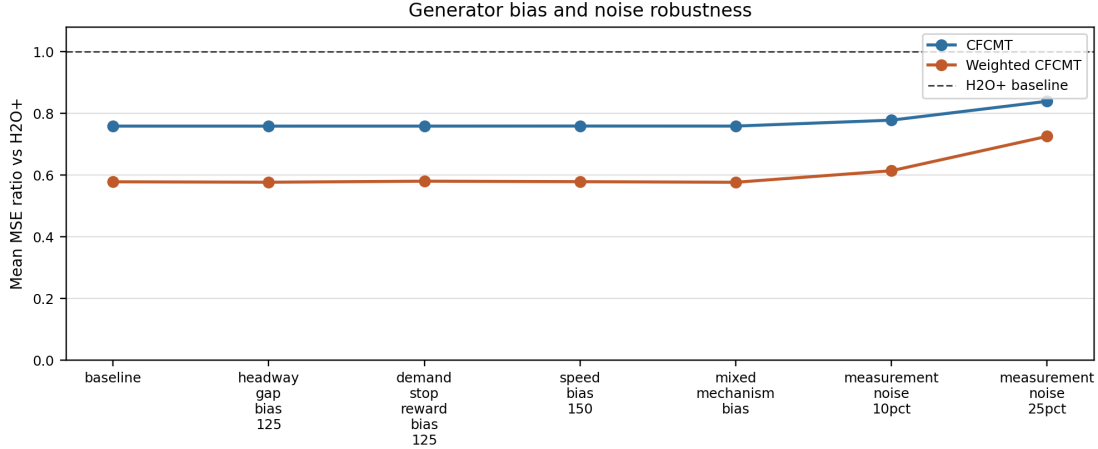


Figure 8: Generator-bias and measurement-noise robustness under strict leave-one-city-out evaluation.

years: the total-MSE ratio is 0.330 on July–December 2021 and 0.323 on August 2016. Unweighted CFCMT also beats the dense baseline, with ratios 0.528 and 0.517. This supports the mechanism-factored transfer claim on real historical APC/AVL observations.

The result should still be read conservatively. The APC-derived no-correction passive baseline has lower total MSE than both residual-transfer models on this real APC slice. This does not mean that a free-running uncalibrated simulator is operationally better; it means that the baseline is given the observed current APC state and that source-only residual corrections can overshoot under real operational noise. The overshoot is concentrated in headway/gap outputs, which dominate the unnormalized total MSE, while CFCMT is better than the passive baseline on several demand, dwell, and speed outputs. Thus the real APC result strengthens the comparison between CFCMT and dense residual transfer, but it also shows that a safe residual gate or small target-city calibration budget is needed before claiming improvement over a state-informed passive baseline.

The few-shot APC sweep confirms this diagnosis. With only 1% of Austin line groups used for target residual calibration, the weighted-CFCMT residual gate improves over the passive baseline in both APC datasets, with ratios 0.970 and 0.957. At 25% target line-group budget, target-only CFCMT reaches the best ratios, 0.893 on July–December 2021 and 0.879 on August 2016. Thus the real APC gap is not evidence against the mechanism-factored representation; it is evidence that source-only residual transfer needs target-domain trust calibration under real APC noise. This is the same qualitative regime targeted by AdaRL/FANsRL-style few-shot adaptation, while the direct AdaRL/FANsRL codebases are environment-specific RL systems rather than drop-in bus residual evaluators.

The chronological time-budget sweep is stricter and closer to a deployment story. One hour of target history is too small: only 2 and 18 calibration transitions are available in the two APC slices, and the residual gate remains above the passive baseline. At six hours, the residual gate reaches parity or better in both datasets, with ratios 1.000 and 0.961. At one day, source-plus-target refitting is below the passive baseline in both datasets, with ratios 0.943 and 0.938. At one week, target-only mechanism-factored adaptation is best, with ratios 0.893 and 0.876. This supports the intended target-offline-adaptation claim: a small but nontrivial amount of passive target history is enough to calibrate mechanism trust, whereas the pure source-only model should not be deployed without such a gate.

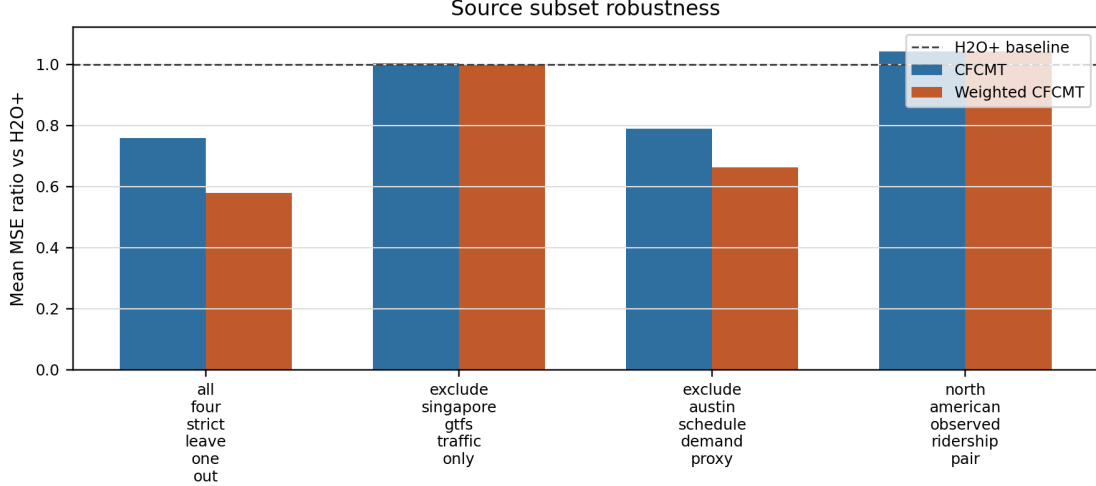


Figure 9: Source-subset robustness. Removing data-source families reveals where the claim becomes weak.

## 6.9 Auxiliary sampled rollout

The sampled BusSimEnv rollout is not primary evidence, but it checks whether the learned one-step action preferences are grossly inconsistent with executable line environments. We expand this diagnostic to 80 route episodes per policy across all four city bundles and add a threshold equalization baseline. The result is mixed and weaker than the one-step policy proxy. On raw mean reward, the dense H2O<sup>+</sup>-style policy is best, with  $-615.8$  versus  $-644.9$  for weighted CFCMT. However, raw rewards have different city scales. After averaging city-level ratios, weighted CFCMT has lower headway error than H2O<sup>+</sup>, with ratio 0.960, and a similar bunching rate, with ratio 0.979. The city-stratified paired bootstrap gives a 95% confidence interval of  $[0.898, 1.008]$  for the weighted-CFCMT/H2O<sup>+</sup> headway-error ratio and  $[0.877, 1.052]$  for the bunching-rate ratio. The city-level pattern remains uneven: H2O<sup>+</sup> wins clearly on Austin reward, while weighted CFCMT improves headway or reward metrics on Halifax, MBTA, and Singapore. Thus the expanded rollout should be read as a boundary condition, not as control superiority evidence: the dynamics-transfer result is strong, but closed-loop policy performance remains metric- and environment-sensitive.

We also add a libsumo full-line counterfactual diagnostic after inserting explicit bus-stop dwell points into generated SUMO scenarios. The executable scenarios contain all validated lines in the four city bundles: 234 Austin, 176 Halifax, 940 MBTA, and 789 Singapore lines, for 58116 generated bus-stop points. The rollout uses a four-hour window, at most eight departures per line, the same  $\{0, 15, 30, 45, 60\}$  second action grid for residual fitting and control, and excludes first-arrival warm-up events whose preceding headway is unobserved. Across the reported policies, the diagnostic evaluates roughly 9.6–10.0 thousand repeated-headway stop decisions after discarding about 7.0–8.0 thousand warm-up stop events. The unconstrained learned residual controllers remain weak in closed loop: dense H2O<sup>+</sup>-MPC has mean headway error 685.55, no holding has 683.84, and weighted CFCMT-MPC improves to 645.35 but still under-holds severe bunching cases. Classical holding is much stronger on this generated topology: threshold equalization reaches 531.35, and a retuned Daganzo rule with  $\alpha = 0.8$  reaches 529.64. The best current executable controller is a target-summary gated Daganzo prior. It uses only target route/timetable summaries to choose a conservative or aggressive Daganzo gain and reaches mean headway error 528.63, reward  $-529.98$ ,

and bunching rate 0.340. Against dense H2O<sup>+</sup>-MPC, the city-stratified paired line bootstrap gives this target-summary prior a reward delta of 155.75 with interval [135.14, 184.26], headway-error ratio 0.660 with interval [0.625, 0.692], and bunching ratio 0.650 with interval [0.593, 0.698]. This diagnostic changes the control-facing interpretation. The mechanism-factored world model improves over dense learned residual MPC, but the closed-loop win over classical rules comes from contextualizing a physically meaningful control prior, not from free-form residual MPC. The generated SUMO scenarios are still not field-calibrated counterfactuals, so the result is best read as a design constraint for the next controller: CFCMT should instantiate target mechanism state and gate a safe holding prior before allowing learned residual actions.

## 6.10 RESCO traffic-signal stress test

The traffic-signal stress test evaluates whether the same mechanism-transfer idea remains useful in a counterfactual control setting where strong classical rules are available. We use eight RESCO SUMO scenarios, three matched random seeds, 600-second closed-loop rollouts, and the original SUMO `tlLogic` green phases as the action set. We also include a generated native SUMO actuated-signal baseline that preserves the same network and route demand but lets SUMO’s built-in actuated controller run without learned phase overrides. The reported metric is mean queue averaged over target scenarios and seeds, with throughput ratio as a secondary check.

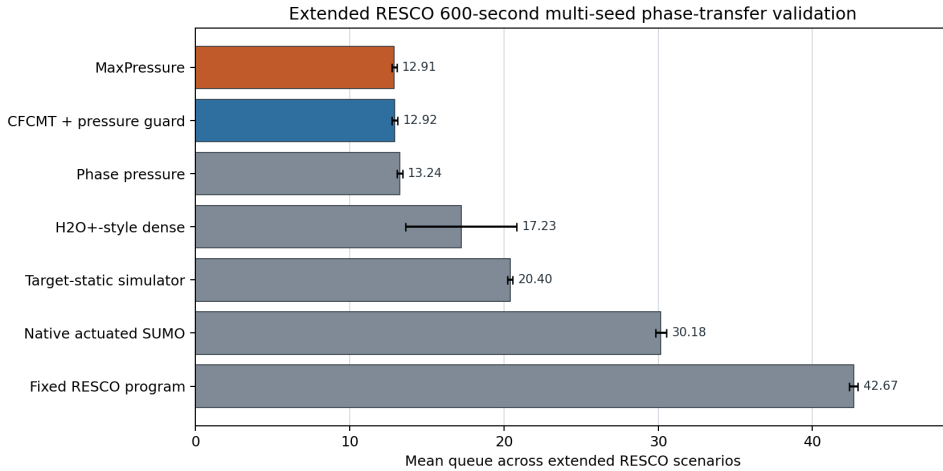


Figure 10: Extended RESCO multi-seed phase-transfer validation. Bars show mean queue across eight held-out RESCO target scenarios and three matched seeds; error bars show seed-level standard deviation of mean queue.

| Policy                  | Mean queue | Seed std | P90 queue | Trip time | Trip wait | Time loss | Throughput |
|-------------------------|------------|----------|-----------|-----------|-----------|-----------|------------|
| MaxPressure             | 12.9068    | 0.1546   | 19.5279   | 87.5891   | 8.0605    | 24.5171   | 0.8267     |
| CFCMT + pressure guard  | 12.9180    | 0.1919   | 19.5973   | 87.4863   | 8.0114    | 24.4127   | 0.8256     |
| Phase pressure          | 13.2446    | 0.1816   | 19.3911   | 86.5371   | 7.4544    | 23.4628   | 0.8199     |
| H2O+-style dense        | 17.2279    | 3.6084   | 28.9325   | 86.6585   | 8.5768    | 24.5488   | 0.7653     |
| Target-static simulator | 20.4023    | 0.1758   | 32.5538   | 97.9633   | 19.9032   | 35.3928   | 0.7429     |
| Native actuated SUMO    | 30.1825    | 0.3536   | 47.6873   | 119.7813  | 33.0017   | 57.0805   | 0.7580     |
| Fixed RESCO program     | 42.6732    | 0.2709   | 69.6852   | 142.3116  | 54.6333   | 79.8759   | 0.7127     |

Table 9: Extended RESCO traffic-signal stress test. Trip time, trip wait, and time loss are official SUMO `tripinfo` metrics over completed vehicles and should be read together with throughput; lower queue and higher throughput are primary.

Figure 10 and table 9 show the main result. MaxPressure has mean queue 12.907, while the pressure-guarded CFCMT phase controller has mean queue 12.918. Their official `tripinfo` metrics are essentially tied: CFCMT has trip time 87.486, trip wait 8.011, and time loss 24.413, compared with 87.589, 8.061, and 24.517 for MaxPressure. The next strongest classical row is phase pressure with mean queue 13.245; it has lower trip time, wait, and time loss, but also lower throughput, so these trip metrics must be interpreted with queue and throughput rather than in isolation. The dense H2O<sup>+</sup>-style phase residual, target-static simulator MPC, generated native actuated SUMO baseline, and fixed RESCO programs are weaker, with mean queues 17.228, 20.402, 30.183, and 42.673, respectively. The paired seed-target bootstrap for guarded CFCMT versus MaxPressure has mean delta 0.011 with 95% interval  $[-0.046, 0.079]$ , so the correct conclusion is near-parity with MaxPressure rather than a significant win. Against phase pressure the mean delta is  $-0.327$  but the interval  $[-0.759, 0.023]$  still crosses zero. Against dense H2O<sup>+</sup>-style residual transfer, target-static simulator MPC, native actuated SUMO, and fixed programs, the intervals favor guarded CFCMT: the mean queue deltas are  $-4.310$ ,  $-7.484$ ,  $-17.265$ , and  $-29.755$ , respectively. Per-target winners remain mixed, which is important for claim scope: several regular or sparse networks are best handled by MaxPressure or a simulator/pressure variant, while the guarded CFCMT controller avoids the large negative transfer seen in dense residual and simulator-only MPC.

The 3600-second long-horizon check in Table 41 is stricter and exposes a boundary. Phase pressure is best at mean queue 32.234 and also has the best trip time, 104.229. MaxPressure and guarded CFCMT rise to mean queues 42.787 and 43.511, with trip times 106.918 and 108.633. Guarded CFCMT still beats dense H2O<sup>+</sup>-style residual transfer, native actuated SUMO, and fixed programs, with mean queue deltas  $-5.995$ ,  $-14.836$ , and  $-41.098$  under the paired seed-target bootstrap, but it is significantly worse than phase pressure on this horizon. Thus the stress test supports guarded transfer as a safe residual design, not as a long-horizon replacement for a well-tuned pressure rule.

This result changes the control-facing message. The evidence is not that CFCMT universally dominates classical traffic control. It is that a mechanism-transfer model can be coupled with a source-only pressure guard to reach MaxPressure-level closed-loop safety on a broad RESCO subset while substantially improving over dense residual transfer and simulator-only MPC under the same phase-transfer protocol.

**Native-protocol RL cross-checks.** Tables 39 and 40 report full-budget external RL runner results. The official RESCO adapter completed 42 supported runs with zero failures: IDQN on all eight extended scenarios and upstream-supported MPLight on six scenarios, while the known upstream MPLight crashes on `cologne8` and `ingolstadt21` are excluded rather than patched into a different algorithm. The patched LibSignal adapter completed 36/36 SUMO runs with zero failures across DQN, PressLight, FRAP, and MPLight on `sumo1x1`, `sumo1x3`, and `sumo4x4`. These tables strengthen the experimental audit trail, but they should not be read as direct wins or losses against CFCMT: RESCO and LibSignal use their own observations, rewards, action timing, and metrics. The same-protocol conclusion remains the one above: guarded CFCMT improves over dense residual transfer and simulator-only MPC and reaches pressure-rule-level safety on the short RESCO stress test, while phase pressure is the stronger long-horizon classical controller.

## 7 Discussion

**Why factored residuals help.** The dense residual baseline has enough capacity to fit cross terms in source data, but the same flexibility can mix city-specific demand, speed, and headway

effects. CFCMT restricts each output to a smaller mechanism-specific parent set. This constraint appears helpful in the strict cross-city setting: it reduces total MSE on all four targets and makes source weighting more effective. The formal risk decomposition in the method section explains this as a bias-control tradeoff: mechanism-wise models reduce cross-mechanism mixing when transfer shifts are localized, while dense residuals may still be competitive when the selected sparse features are the main source of gain. These experiments support mechanism-factored transfer as a useful inductive bias; they do not establish formal causal identification or invariance under intervention.

**What the source weighting and source selection do and do not show.** Similarity weighting improves the mean strict leave-one-out ratio from 0.759 to 0.578 and wins over unweighted CFCMT on three of four targets. The Singapore target is the exception, where weighting is slightly worse than unweighted CFCMT. The full-module feature analysis makes the boundary sharper: a source-only feature-set selector chooses simulator-output summaries for most splits, observation summaries for the duplicated Singapore target, and never selects local-only or uniform weighting. Thus the useful signal in this diagnostic is not the weak static local graph itself, but the target’s unlabeled transition-summary statistics. An expanded NYC/SEPTA/MBTA GTFS-RT occupancy snapshot check gives the same warning: static local features have cross-city mean ratio 1.011 against no-local features and the selector falls back to no-local, while within-city temporal prediction is essentially tied with mean ratio 0.997. The guarded full-module ensemble makes this boundary more explicit. It helps when the unlabeled target summary identifies a dominant nearby source, as in MBTA and Austin, but the guard falls back to rigid CFCMT when source similarity is diffuse, as in Halifax. The same-information dense  $\text{H}_2\text{O}^+$  source ensemble also improves substantially, so the target-summary signal is genuinely useful; the remaining gap favors mechanism-factored residuals over dense residual mixing. This is not target-label calibration: the target labels remain unseen, but the method is not a pure source-only setting because it uses unlabeled target summaries for source selection.

**Zero-shot is a lower bound, not the intended deployment endpoint.** The single-source transfer matrix is useful precisely because it is uneven. It shows that a mechanism model trained on one city can transfer in some directions, but it also exposes source-target pairs where one-city history is not enough. The stronger and more realistic claim is a target-information ladder: multi-source target-label-free transfer gives the main zero-target-label result, and few-shot target offline adaptation uses a small amount of passive target history to tune mechanism trust without claiming online deployment. This framing matches practical transit deployment better than a pure zero-shot claim, because agencies normally have at least some archived schedules, counts, AVL traces, or APC samples before changing a control policy. The Austin APC time-budget sweep makes this concrete: one hour is too little to trust, six hours is enough for the residual gate to reach parity or better in both slices, and one day to one week gives a clearer target-adaptation benefit. The target-route baseline comparison also changes the preferred few-shot implementation. Full mechanism refitting is not the safest way to use a small target budget; a low-dimensional target context-bias adapter is consistently stronger on average. This is consistent with the causal-mechanism view: passive target history should first update mechanism state or trust, not overwrite all transferred mechanism parameters.

**Policy evidence remains secondary.** The one-step policy proxy and sampled BusSimEnv roll-out are useful checks, but they do not replace a closed-loop operational benchmark. The expanded sampled rollout is especially important because it weakens the earlier parity story: the raw aggre-

gate favors  $H2O^+$ , while city-normalized headway and bunching metrics favor or nearly tie weighted CFCMT. That correction is useful rather than damaging; it keeps the policy claim bounded and prevents the paper from overstating a dynamics-transfer result as operational dominance. The full-line libsumo diagnostic strengthens the control story only after adding a safe target-summary prior. Unconstrained learned residual MPC improves over dense learned MPC but remains far weaker than classical holding rules in severe bunching states. By contrast, a target-summary gated Daganzo prior, which uses static route and schedule summaries to choose the holding gain, slightly improves over the best fixed Daganzo gain in the four-city diagnostic. This suggests that the next controller should not be a free-form residual action selector. It should instantiate a target mechanism state from route, schedule, demand, and AVL/APC summaries, use that state to gate a physically meaningful holding prior, and allow learned residual actions only behind a trust gate. The strongest claim should therefore remain dynamics transfer plus bounded policy improvement, not field-ready policy dominance.

**What the RESCO stress test changes.** The RESCO result gives a cleaner counterfactual-control test than passive bus logs because each candidate signal phase can be rolled out in SUMO. It also imposes a harder baseline: MaxPressure is close to optimal in several regular network regimes. The pressure-guarded CFCMT controller should therefore be interpreted as a safe transfer design, not as a universal replacement for pressure control. On the extended RESCO subset it reaches MaxPressure-level mean queue, is close to phase pressure with a non-significant mean advantage, and significantly improves over dense residual transfer, simulator-only MPC, spillback pressure, and fixed programs. The 3600-second check is even more conservative: phase pressure becomes clearly best, while guarded CFCMT remains closer to MaxPressure and still better than dense residual transfer and fixed programs. The non-significant short-horizon gaps against MaxPressure and phase pressure, together with the long-horizon phase-pressure win, are scientifically useful: they show that the method must preserve physically meaningful priors when a classical rule is already near the ceiling. This is also the most defensible direction for future policy work in bus holding, where a gated Daganzo or threshold prior is currently stronger than unconstrained learned residual MPC.

**Why the RL leaderboard tables are separated.** The official RESCO and LibSignal full-budget runs are included to address a common traffic-control review question: whether the proposed controller was compared against standard RL baselines rather than only hand-coded rules. They also expose a practical point. On these benchmarks, full-budget RL training is expensive and implementation-specific; for example, LibSignal FRAP on `sumo4x4` is orders of magnitude slower than small-grid runs even after cProfile-driven simulator and parallelization fixes. More importantly, these libraries do not use the same observation representation, reward, control interval, or metric aggregation as the CFCMT phase-MPC stress test. Putting all rows into one ranked table would therefore create a false sense of protocol equivalence. The correct use is as a native-protocol leaderboard cross-check in the appendix, while the main causal-transfer comparison remains the same-protocol phase-MPC table.

**Static-derived validation is not operational validation.** The central limitation is target construction. The experiments use full-route open static data and deterministic transition rules to create paired simulator/target transitions. This is valuable for reproducibility and reviewer scrutiny, but it is not the same as evaluating counterfactual holding actions on AVL/APC trajectories or a calibrated live SUMO rollout. The Austin raw APC/AVL slice partially addresses the trajectory-



ground-truth concern for passive action-0 dynamics, but it still cannot identify how unobserved holding actions would have changed passenger and headway outcomes. True counterfactual holding labels would require an intervention design: randomized or rule-threshold hold/no-hold variation in the field, a natural experiment that changes holding behavior while leaving comparable states, or a simulator calibrated and validated well enough to serve as the counterfactual environment. The paper should therefore avoid claims such as “CFCMT is better on real bus systems”. A precise claim is: under open static-data-derived cross-city validation with no or few target labels, and under passive real APC validation with few-shot adaptation, mechanism-factored residual transfer with unlabeled target-summary source selection is more robust than dense residual correction.

**How to strengthen the next version.** The most valuable addition would be counterfactual or interventional validation on one city with vehicle-level AVL/APC trajectories, even if only for a subset of routes. The second priority is a calibrated or independently validated closed-loop simulator, so the target-summary-gated prior and any learned residual override can be tested against real or validated counterfactual outcomes. The third priority is to add more cities, because city-level generalization is ultimately limited by the number of held-out cities.

## 8 Conclusion

We presented CFCMT, a mechanism-factored residual transfer approach for bus holding control under cross-city open-transit validation. Compared with a dense  $H_2O^+$ -style residual baseline, CFCMT improves strict leave-one-city-out dynamics on all four target cities, and source-city similarity weighting further improves the mean MSE ratio. The target-information ladder clarifies the deployment regimes: single-source zero-shot transfer is mixed, multi-source target-label-free transfer is the strongest zero-target-label result, and few-shot target offline adaptation is the practical setting when a new city has limited passive history. The few-shot comparison against AdaRL/FANsRL-style same-information adapters further shows that small target histories are best used to estimate a low-dimensional mechanism context bias rather than to refit every transferred mechanism parameter. The full-module guarded ensemble adds a target-label-free source-selection layer: it uses source-fold validation to choose unlabeled target summaries, then selects source residual experts and improves the configured selector-stress validation while falling back to rigid CFCMT when no dominant source is identified. One-step policy-proxy evaluation also favors CFCMT on most reward/regret and headway metrics and on all bunching-rate comparisons. At the same time, subset robustness shows that the advantage depends on source diversity, and generator robustness shows that not every perturbation preserves all-target wins. The Austin raw APC/AVL check adds a sharper boundary condition: source-only transfer is better than dense residual transfer but can still overshoot a strong state-informed passive baseline, while few-shot residual gating closes that gap. The chronological APC budget curve further shows that this is not merely a route-random artifact: six hours of passive target history is enough for the residual gate to reach parity or better in both Austin slices, and one week gives the best target-only mechanism adaptation.

The traffic-signal stress test gives a separate counterfactual-control check. On the short RESCO phase-transfer protocol, pressure-guarded CFCMT reaches MaxPressure-level mean queue and improves over dense residual transfer, simulator-only MPC, native actuated SUMO, and fixed programs under the same protocol. The longer 3600-second check is deliberately more conservative: phase pressure becomes the strongest classical rule, while guarded CFCMT remains closer to the pressure-rule safety region than dense residual transfer. The official RESCO and patched LibSignal full-budget RL tables further show that standard RL runners were connected and executed, but

they are reported as native-protocol leaderboard cross-checks rather than as direct same-protocol wins or losses against CFCMT.

The resulting paper claim is intentionally bounded. CFCMT is a calibration-light causal mechanism transfer layer that improves dense residual correction and simulator-only transfer under cross-network validation, can use passive target history through low-dimensional mechanism trust adaptation, and can reach pressure-rule-level safety when coupled with physically meaningful guards. It is not a claim that CFCMT universally dominates classical pressure control or that passive AVL/APC logs identify unobserved holding-action counterfactuals. Operational deployment claims still require counterfactual control evidence from a validated simulator, credible intervention design, or field validation.

## A Additional Tables

### A.1 Target-construction audit

| city              | demand_evidence               | traffic_evidence     | lines | transitions | uncal_mse | line_mse_p25 | line_mse_p50 | line_mse_p75 |
|-------------------|-------------------------------|----------------------|-------|-------------|-----------|--------------|--------------|--------------|
| Singapore         | observed stop OD/PV monthly   | observed speed bands | 789   | 1243488     | 2256.5339 | 220.173      | 324.7508     | 541.4336     |
| Austin / CapMetro | schedule proxy                | schedule derived     | 234   | 310272      | 1301.3027 | 242.2482     | 615.8957     | 3043.8061    |
| Halifax Transit   | route APC apportioned         | schedule derived     | 176   | 266448      | 1111.9223 | 199.2164     | 620.2908     | 1614.3016    |
| MBTA              | stop board/alight apportioned | schedule derived     | 940   | 969360      | 1216.3716 | 180.9625     | 459.6093     | 1040.0624    |

Table 10: Target-construction audit and route-level nontriviality check.

### A.2 Target offline adaptation budget

| budget | splits | mean_no_cal_ratio | mean_cal_ratio | no_cal_wins | cal_wins | mean_calib_lines | oracle |
|--------|--------|-------------------|----------------|-------------|----------|------------------|--------|
| 0.0    | 4      | 0.5782            | 0.5782         | 4           | 4        | 0.0              | False  |
| 0.01   | 4      | 0.5807            | 0.5801         | 3           | 3        | 3.5              | False  |
| 0.05   | 4      | 0.5863            | 0.5831         | 3           | 3        | 23.0             | False  |
| 0.1    | 4      | 0.6165            | 0.6094         | 3           | 3        | 50.5             | False  |
| 0.25   | 4      | 0.6772            | 0.6596         | 3           | 3        | 132.75           | False  |
| 1.0    | 4      | 0.7872            | 0.7242         | 3           | 3        | 534.75           | True   |

Table 11: Target-route offline-adaptation budget summary. Budget 1.0 is an in-sample oracle reference.

### A.3 Per-mechanism errors

| mechanism          | targets | weighted_wins | mean_cfcmt_ratio | mean_weighted_ratio |
|--------------------|---------|---------------|------------------|---------------------|
| demand_stop_reward | 4       | 4             | 0.8197           | 0.7706              |
| headway_gap        | 4       | 3             | 0.7586           | 0.5743              |
| speed              | 4       | 4             | 0.8203           | 0.6684              |

Table 12: Per-mechanism strict leave-one-city-out error ratios.

#### A.4 Mechanism and capacity ablations

| family                          | wins_vs_h2oplus | mean_ratio_vs_h2oplus |
|---------------------------------|-----------------|-----------------------|
| cfcmt_shared_sparse             | 4               | 0.7502                |
| cfcmt_no_action_interaction     | 4               | 0.7583                |
| cfcmt_full                      | 4               | 0.7587                |
| dense_matched_sparse            | 3               | 0.7716                |
| dense_all_parents_no_pairwise   | 3               | 0.9332                |
| cfcmt_action_time_only          | 1               | 7.3401                |
| cfcmt_random_mechanism_grouping | 1               | 7.5744                |

Table 13: Mechanism and capacity ablations. Lower mean ratios indicate better transfer relative to H2O<sup>+</sup>.

#### A.5 Route-bootstrap confidence intervals

| target_city       | lines | cfcmt_ratio | cfcmt_ratio.ci95 | weighted_ratio | weighted_ratio.ci95 |
|-------------------|-------|-------------|------------------|----------------|---------------------|
| Singapore         | 789   | 0.9964      | [0.991, 1.000]   | 0.9986         | [0.993, 1.002]      |
| Austin / CapMetro | 234   | 0.7375      | [0.670, 0.817]   | 0.3985         | [0.354, 0.447]      |
| Halifax Transit   | 176   | 0.7166      | [0.529, 0.871]   | 0.5638         | [0.310, 0.783]      |
| MBTA              | 940   | 0.5844      | [0.520, 0.643]   | 0.352          | [0.257, 0.438]      |

Table 14: Route-level bootstrap confidence intervals under strict leave-one-city-out dynamics validation.

#### A.6 External real APC/AVL validation

| dataset                                    | transitions | routes | stops | trips  | date_min   | date_max   | h2oplus_vs_passive | weighted_vs_passive | cfcmt_ratio | weighted_ratio |
|--------------------------------------------|-------------|--------|-------|--------|------------|------------|--------------------|---------------------|-------------|----------------|
| CapMetro APC Raw July 2021 - December 2021 | 5771472     | 68     | 2315  | 421255 | 2021-07-01 | 2021-12-30 | 10.1614            | 3.352               | 0.5281      | 0.3299         |
| CapMetro APC Raw August 2016               | 5796060     | 72     | 2576  | 220011 | 2016-08-21 | 2017-01-07 | 10.3665            | 3.3452              | 0.5173      | 0.3227         |

Table 15: External Austin raw APC/AVL validation. The observed data provide passive action-0 stop-event transitions, not counterfactual holding actions.

| dataset                                    |      | budget | cal_lines | cal_transitions | source_only_vs_passive | source_plus_target_vs_passive | target_only_vs_passive | residual_gate_vs_passive |
|--------------------------------------------|------|--------|-----------|-----------------|------------------------|-------------------------------|------------------------|--------------------------|
| CapMetro APC Raw July 2021 - December 2021 | 0.0  | 0      | 0         | 3.352           | 3.352                  | -                             | 3.352                  | 3.352                    |
| CapMetro APC Raw July 2021 - December 2021 | 0.01 | 1      | 745       | 3.352           | 0.9803                 | 1.0662                        | 0.97                   | 0.97                     |
| CapMetro APC Raw July 2021 - December 2021 | 0.05 | 8      | 293425    | 3.3466          | 0.93                   | 0.8982                        | 0.9701                 | 0.9701                   |
| CapMetro APC Raw July 2021 - December 2021 | 0.1  | 12     | 650668    | 3.3437          | 0.9207                 | 0.9087                        | 0.9699                 | 0.9699                   |
| CapMetro APC Raw July 2021 - December 2021 | 0.25 | 24     | 1537062   | 3.3381          | 0.911                  | 0.8933                        | 0.9699                 | 0.9699                   |
| CapMetro APC Raw August 2016               | 0.0  | 0      | 0         | 3.3452          | 3.3452                 | -                             | 3.3452                 | 3.3452                   |
| CapMetro APC Raw August 2016               | 0.01 | 1      | 5088      | 3.3452          | 0.9515                 | 0.9765                        | 0.9567                 | 0.9567                   |
| CapMetro APC Raw August 2016               | 0.05 | 8      | 512496    | 3.3556          | 0.9028                 | 0.8749                        | 0.9551                 | 0.9551                   |
| CapMetro APC Raw August 2016               | 0.1  | 14     | 922153    | 3.3472          | 0.8963                 | 0.8784                        | 0.9549                 | 0.9549                   |
| CapMetro APC Raw August 2016               | 0.25 | 26     | 1620309   | 3.3277          | 0.8903                 | 0.879                         | 0.9556                 | 0.9556                   |

Table 16: Few-shot Austin raw APC/AVL adaptation. Ratios are total MSE relative to the state-informed passive no-correction baseline on held-out Austin line groups.

| dataset                                    | hours | cal_lines | cal_transitions | eval_transitions | source_only_vs_passive | source_plus_target_vs_passive | target_only_vs_passive | residual_gate_vs_passive | bias_adapter_vs_passive |
|--------------------------------------------|-------|-----------|-----------------|------------------|------------------------|-------------------------------|------------------------|--------------------------|-------------------------|
| CapMetro APC Raw July 2021 - December 2021 | 0.0   | 0         | 0               | 5771472          | 3.352                  | 3.352                         | -                      | 3.352                    | 3.352                   |
| CapMetro APC Raw July 2021 - December 2021 | 1.0   | 1         | 2               | 5771470          | 3.352                  | 2.1209                        | 3.8632                 | 1.8777                   | 3.9879                  |
| CapMetro APC Raw July 2021 - December 2021 | 6.0   | 30        | 442             | 5771030          | 3.3518                 | 1.1627                        | 3.5578                 | 0.9907                   | 8.4575                  |
| CapMetro APC Raw July 2021 - December 2021 | 24.0  | 76        | 32348           | 5739124          | 3.3521                 | 0.9432                        | 0.9045                 | 0.9702                   | 3.2114                  |
| CapMetro APC Raw July 2021 - December 2021 | 168.0 | 87        | 215429          | 5556043          | 3.3512                 | 0.9341                        | 0.8928                 | 0.9698                   | 3.1809                  |
| CapMetro APC Raw August 2016               | 0.0   | 0         | 0               | 5796060          | 3.3452                 | 3.3452                        | -                      | 3.3452                   | 3.3452                  |
| CapMetro APC Raw August 2016               | 1.0   | 6         | 18              | 5796042          | 3.3452                 | 1.3049                        | 1.2877                 | 1.0133                   | 6.2691                  |
| CapMetro APC Raw August 2016               | 6.0   | 9         | 100             | 5795960          | 3.3452                 | 1.1928                        | 1.2705                 | 0.9611                   | 5.6604                  |
| CapMetro APC Raw August 2016               | 24.0  | 44        | 19116           | 5776944          | 3.3445                 | 0.9384                        | 0.9695                 | 0.9595                   | 3.7053                  |
| CapMetro APC Raw August 2016               | 168.0 | 83        | 282958          | 5513102          | 3.3454                 | 0.9091                        | 0.8756                 | 0.9558                   | 3.1596                  |

Table 17: Chronological few-shot Austin raw APC/AVL adaptation. The first  $N$  hours of target APC history are used for adaptation, and later APC transitions are held out for evaluation. Ratios are total MSE relative to the state-informed passive no-correction baseline.

## A.7 Configured six-split dynamics validation

## A.8 Guarded source-ensemble diagnostics

## A.9 Source-weighting sensitivity

## A.10 Source-size sensitivity and single-source zero-shot transfer

## A.11 Generator robustness

## A.12 Source-subset robustness

## A.13 Auxiliary sampled rollout

## A.14 Native-protocol traffic-signal RL baselines

## A.15 Traffic-signal long-horizon check

| target      | transitions | snapshots | routes | no_local_mse | static_local_mse | static_vs_no | selector | selector_vs_no | temporal_static_vs_no |
|-------------|-------------|-----------|--------|--------------|------------------|--------------|----------|----------------|-----------------------|
| mbta_live   | 1666        | 11        | 63     | 0.0471       | 0.0484           | 1.0279       | no_local | 1.0            | 1.019                 |
| nyc_mta_bus | 52920       | 95        | 241    | 0.1569       | 0.1564           | 0.997        | no_local | 1.0            | 0.9832                |
| septa_bus   | 17040       | 94        | 96     | 0.6399       | 0.6457           | 1.0091       | no_local | 1.0            | 0.9902                |

Table 18: NYC/SEPTA/MBTA GTFS-RT occupancy snapshot validation. The task is one-step occupancy ordinal prediction from public synchronized vehicle-position snapshots; it is a local-graph feasibility diagnostic, not a control-action validation.

| split                                      | target_city       | train_transitions | target_transitions | h2oplus_total_mse | cfcmt_total_mse | cfcmt_vs_h2oplus_ratio | weighted_cfcmt_vs_h2oplus_ratio | weighted_cfcmt_vs_unweighted_ratio |
|--------------------------------------------|-------------------|-------------------|--------------------|-------------------|-----------------|------------------------|---------------------------------|------------------------------------|
| source_open_na_to_singapore                | Singapore         | 1546080           | 1243488            | 2254.8548         | 2246.6693       | 0.9964                 | 0.9986                          | 1.0022                             |
| source_singapore_austin_halifax_to_mbt     | MBTA              | 1820208           | 969360             | 318.9925          | 186.4177        | 0.5844                 | 0.352                           | 0.6022                             |
| leave_one_city_out_all_singapore_ita_all   | Singapore         | 1546080           | 1243488            | 2254.8548         | 2246.6693       | 0.9964                 | 0.9986                          | 1.0022                             |
| leave_one_city_out_all_austin_capmetro_all | Austin / CapMetro | 2479296           | 310272             | 91.9724           | 67.8254         | 0.7375                 | 0.3985                          | 0.5404                             |
| leave_one_city_out_all_halifax_transit_all | Halifax Transit   | 2523120           | 266148             | 97.1175           | 69.5992         | 0.7166                 | 0.5638                          | 0.7867                             |
| leave_one_city_out_all_mbtta_all           | MBTA              | 1820208           | 969360             | 318.9925          | 186.4177        | 0.5844                 | 0.352                           | 0.6022                             |

Table 19: Configured cross-city dynamics splits. This table includes repeated target cities and is treated as appendix evidence rather than the primary independent-sample result.

| component                   | original H2O+ role                            | implemented here                            | deviation / reason                                  |
|-----------------------------|-----------------------------------------------|---------------------------------------------|-----------------------------------------------------|
| offline residual correction | learn simulator-to-data correction            | ridge dense residual on source transitions  | sufficient-statistics surrogate for reproducibility |
| target calibration          | may adapt with target data or online feedback | excluded from primary cross-city fit        | tests target-label-free transfer                    |
| source/target domains       | not explicitly separated in original setting  | source cities train, target city held out   | cross-city protocol requires domain split           |
| policy learning loop        | optimize policy with corrected dynamics       | one-step policy proxy and auxiliary rollout | paper focuses on world-model transfer               |
| same-information variant    | not a separate original component             | source-weighted dense residual ensemble     | fairness check for unlabeled target-summary use     |

Table 20: Baseline-fidelity note for the dense H2O<sup>+</sup>-style residual baselines. The implementation is a reproducible dense residual transfer surrogate, not a full reproduction of the original H2O+ policy-learning stack.

| feature family | unlabeled target information                        | purpose                                            |
|----------------|-----------------------------------------------------|----------------------------------------------------|
| sim_only       | summary statistics of simulator outputs $f_0(s, a)$ | compare generated transition regimes               |
| obs_only       | state/action covariate summaries                    | compare target operating conditions                |
| obs_sim        | state/action plus simulator-output summaries        | combine operating and generated-transition signals |
| local_only     | deterministic local static graph descriptors        | test static local-graph signal alone               |
| full           | obs_only, sim_only, and local_only concatenated     | broad descriptor ablation                          |
| uniform        | no target similarity descriptor                     | no target-summary weighting control                |

Table 21: Candidate unlabeled target-summary feature families used by the source-only feature-set selector.

| method                              | uses_unlabeled_target_summary | uses_target_transition_labels_for_selection | mean_total_mse | mean_ratio_vs_h2oplus | wins_vs_h2oplus |
|-------------------------------------|-------------------------------|---------------------------------------------|----------------|-----------------------|-----------------|
| H2O+ pooled dense residual          | False                         | False                                       | 884.8525       | 1.0                   | 0               |
| H2O+ source-weighted dense residual | True                          | False                                       | 846.4656       | 0.796                 | 5               |
| Rigid CFCMT                         | False                         | False                                       | 851.8929       | 0.7651                | 4               |
| Guarded CFCMT source ensemble       | True                          | False                                       | 836.1148       | 0.7082                | 6               |

Table 22: Same-information baseline for the guarded source-ensemble diagnostic. The dense H2O<sup>+</sup> source ensemble receives the same unlabeled target-summary source weights as the guarded mechanism-factored ensemble.

| method                                                    | result_source             | splits | mean_total_mse | mean_ratio_vs_h2oplus | wins_vs_h2oplus |
|-----------------------------------------------------------|---------------------------|--------|----------------|-----------------------|-----------------|
| H2O+ dense ridge                                          | guarded run               | 6      | 884.8525       | 1.0                   | 0               |
| H2O+ dense source ensemble, same unlabeled target summary | guarded run               | 6      | 846.4656       | 0.796                 | 5               |
| Rigid CFCMT                                               | guarded run               | 6      | 851.8929       | 0.7651                | 4               |
| Rigid-first pooled delta selector                         | non-ensemble selector run | 6      | 844.8108       | 0.7621                | 6               |
| Source-city ensemble selector, unguarded                  | unguarded ensemble run    | 6      | 838.6957       | 0.7446                | 6               |
| Source-city ensemble selector, guarded                    | guarded ensemble run      | 6      | 836.1148       | 0.7082                | 6               |
| Per-mechanism splice selector                             | guarded ensemble run      | 6      | 864.5925       | 0.8487                | 6               |

Table 23: Guarded source-ensemble selector ablation. The guarded row is the reported full-module diagnostic; unguarded and pooled rows isolate which selector components matter.

| target_city     | source_cities                                   | selected_candidate                           | h2oplus_mse | rigid_cfcmt_mse | guarded_mse | guarded_ratio_vs_h2oplus | guarded_ratio_vs_rigid | max_source_weight |
|-----------------|-------------------------------------------------|----------------------------------------------|-------------|-----------------|-------------|--------------------------|------------------------|-------------------|
| singapore_ita   | austin capmetro, halifax transit, mbta          | dense_static_local_delta_alpha_0p75          | 2376.1427   | 2388.7628       | 2367.5165   | 0.9964                   | 0.9911                 | 0.465             |
| austin capmetro | singapore_ita, halifax transit, mbta            | ensemble_dense_static_local_delta_alpha_0p25 | 63.8731     | 47.6396         | 40.7936     | 0.6387                   | 0.8563                 | 0.5905            |
| halifax transit | singapore_ita, austin capmetro, mbta            | rigid_cfcmt                                  | 73.5689     | 53.1982         | 53.1982     | 0.7231                   | 1.0                    | 0.3749            |
| mbta            | singapore_ita, austin capmetro, halifax transit | ensemble_dense_static_local_delta_alpha_0p50 | 209.6939    | 116.4969        | 93.8321     | 0.4475                   | 0.8054                 | 0.5909            |
| mean            |                                                 |                                              | 680.8197    | 651.5244        | 638.8351    | 0.7014                   | 0.9132                 |                   |

Table 24: Strict four-target subset of the guarded source-ensemble diagnostic. The mean row averages only unique leave-one-city-out targets.

| ensemble_min_win_rate | ensemble_min_max_source_weight | splits | mean_total_mse | mean_ratio_vs_h2oplus | wins_vs_h2oplus | ensemble_selected_splits | rigid_fallback_splits |
|-----------------------|--------------------------------|--------|----------------|-----------------------|-----------------|--------------------------|-----------------------|
| 0.6667                | 0.4                            | 6      | 836.1148       | 0.7082                | 6               | 3                        | 1                     |
| 0.6667                | 0.5                            | 6      | 836.1148       | 0.7082                | 6               | 3                        | 1                     |
| 0.8                   | 0.4                            | 6      | 836.1148       | 0.7082                | 6               | 3                        | 1                     |
| 0.8                   | 0.5                            | 6      | 836.1148       | 0.7082                | 6               | 3                        | 1                     |
| 1.0                   | 0.4                            | 6      | 837.2558       | 0.7261                | 6               | 2                        | 2                     |
| 1.0                   | 0.5                            | 6      | 837.2558       | 0.7261                | 6               | 2                        | 2                     |
| 0.6667                | 0.0                            | 6      | 837.4242       | 0.726                 | 6               | 4                        | 0                     |
| 0.8                   | 0.0                            | 6      | 837.4242       | 0.726                 | 6               | 4                        | 0                     |
| 1.0                   | 0.0                            | 6      | 838.5652       | 0.7439                | 6               | 3                        | 1                     |
| 0.6667                | 0.6                            | 6      | 844.8108       | 0.7621                | 6               | 0                        | 4                     |
| 0.8                   | 0.6                            | 6      | 844.8108       | 0.7621                | 6               | 0                        | 4                     |
| 1.0                   | 0.6                            | 6      | 844.8108       | 0.7621                | 6               | 0                        | 4                     |

Table 25: Guarded ensemble threshold sensitivity. The reported default uses ensemble minimum win rate 0.8 and minimum maximum-source weight 0.5.

| configuration               | result_source          | selector_weighting | ensemble_weighting | similarity_feature_set | mean_total_mse | mean_ratio_vs_h2oplus | wins_vs_h2oplus | ensemble_selected_splits | rigid_fallback_splits |
|-----------------------------|------------------------|--------------------|--------------------|------------------------|----------------|-----------------------|-----------------|--------------------------|-----------------------|
| reported guarded (sim_only) | reported guarded       | similarity         | similarity         | sim_only               | 836.1148       | 0.7082                | 6               | 3                        | 1                     |
| obs_sim                     | feature ablation rerun | similarity         | similarity         | obs_sim                | 836.2455       | 0.7092                | 6               | 3                        | 1                     |
| full                        | feature ablation rerun | similarity         | similarity         | full                   | 836.2719       | 0.7094                | 6               | 3                        | 1                     |
| obs_only                    | feature ablation rerun | similarity         | similarity         | obs_only               | 836.3417       | 0.71                  | 6               | 3                        | 1                     |
| local_only                  | feature ablation rerun | similarity         | similarity         | local_only             | 844.8108       | 0.7621                | 6               | 0                        | 4                     |
| uniform                     | feature ablation rerun | uniform            | uniform            | full                   | 846.2431       | 0.7627                | 6               | 0                        | 4                     |

Table 26: Similarity-feature ablation for the guarded source ensemble. All rows use the same source-only fitting and selector protocol unless otherwise noted.

| split                                       | selected_feature_set | source_fold_score | selected_candidate                           | h2oplus_mse | guarded_mse | guarded_ratio_vs_h2oplus |
|---------------------------------------------|----------------------|-------------------|----------------------------------------------|-------------|-------------|--------------------------|
| source_open_na_to_singapore                 | obs_only             | 0.8941            | dense_static_local_delta_alpha_0p75          | 2376.1427   | 2367.5165   | 0.9964                   |
| source_singapore_austin_halifax_to_mbta     | sim_only             | 0.6418            | ensemble_dense_static_local_delta_alpha_0p50 | 209.6939    | 93.8321     | 0.4475                   |
| leave_one_city_out_all::singapore_ita_all   | obs_only             | 0.8941            | dense_static_local_delta_alpha_0p75          | 2376.1427   | 2367.5165   | 0.9964                   |
| leave_one_city_out_all::austin_capmetro_all | sim_only             | 0.8808            | ensemble_dense_static_local_delta_alpha_0p25 | 63.8731     | 40.7936     | 0.6387                   |
| leave_one_city_out_all::halifax_transit_all | sim_only             | 1.0               | rigid_cfcmt                                  | 73.5689     | 53.1982     | 0.7231                   |
| leave_one_city_out_all::mbta_all            | sim_only             | 0.6418            | ensemble_dense_static_local_delta_alpha_0p50 | 209.6939    | 93.8321     | 0.4475                   |
| mean                                        |                      | 0.8254            |                                              | 884.8525    | 836.1148    | 0.7082                   |

Table 27: Source-only feature-set selector for the guarded source ensemble. The feature set is chosen from source-fold scores before target evaluation.

| candidate_family                  | target_oracle_best_method                                | mean_total_mse | mean_ratio_vs_h2oplus | wins_vs_h2oplus |
|-----------------------------------|----------------------------------------------------------|----------------|-----------------------|-----------------|
| ensemble_dense_static_local_delta | rigid_first_ensemble_dense_static_local_delta_alpha_0p75 | 837.5112       | 0.7067                | 6               |
| ensemble_static_local_delta       | rigid_first_ensemble_static_local_delta_alpha_0p75       | 838.5471       | 0.7363                | 6               |
| ensemble_dense_delta              | rigid_first_ensemble_dense_delta_alpha_0p50              | 846.3213       | 0.735                 | 4               |
| static_local_delta                | rigid_first_static_local_delta_alpha_0p25                | 848.4001       | 0.7737                | 4               |
| dense_static_local_delta          | rigid_first_dense_static_local_delta_alpha_0p25          | 851.3022       | 0.7533                | 4               |
| dense_delta                       | rigid_first_dense_delta_alpha_0p25                       | 852.9865       | 0.7597                | 4               |

Table 28: Target-oracle diagnostic over rigid-first candidate families. This table is diagnostic only because it chooses the best candidate family by target labels after evaluation.

| split                                       | selected_candidate                           | selected_mse | candidate_oracle_method                           | candidate_oracle_mse | selected_over_oracle | non_ensemble_oracle_method                      | non_ensemble_oracle_mse | selected_over_non_ensemble_oracle |
|---------------------------------------------|----------------------------------------------|--------------|---------------------------------------------------|----------------------|----------------------|-------------------------------------------------|-------------------------|-----------------------------------|
| source_open_na_to_singapore                 | dense_static_local_delta_alpha_0p75          | 2367.5165    | rigid_first_static_local_delta_alpha_0p0          | 2359.1022            | 1.0036               | rigid_first_static_local_delta_alpha_0p0        | 2359.1022               | 1.0036                            |
| source_singapore_austin_halifax_to_mbta     | ensemble_dense_static_local_delta_alpha_0p50 | 93.8321      | rigid_first_ensemble_static_local_delta_alpha_0p0 | 86.8407              | 1.0805               | rigid_first_static_local_delta_alpha_0p50       | 110.063                 | 0.8525                            |
| leave_one_city_out_all::singapore_ita_all   | dense_static_local_delta_alpha_0p75          | 2367.5165    | rigid_first_static_local_delta_alpha_0p0          | 2359.1022            | 1.0036               | rigid_first_static_local_delta_alpha_0p0        | 2359.1022               | 1.0036                            |
| leave_one_city_out_all::austin_capmetro_all | ensemble_dense_static_local_delta_alpha_0p25 | 40.7936      | rigid_first_ensemble_dense_delta_alpha_0p0        | 32.0227              | 1.2739               | rigid_first_static_local_delta_alpha_0p25       | 43.3178                 | 0.9417                            |
| leave_one_city_out_all::halifax_transit_all | rigid_cfcmt                                  | 53.1982      | rigid_first_dense_static_local_delta_alpha_0p50   | 41.0807              | 1.295                | rigid_first_dense_static_local_delta_alpha_0p50 | 41.0807                 | 1.295                             |
| leave_one_city_out_all::mbta_all            | ensemble_dense_static_local_delta_alpha_0p50 | 93.8321      | rigid_first_ensemble_static_local_delta_alpha_0p0 | 86.8407              | 1.0805               | rigid_first_static_local_delta_alpha_0p50       | 110.063                 | 0.8525                            |
| mean                                        |                                              | 836.1148     |                                                   | 827.4982             | 1.1228               |                                                 | 837.1215                | 0.9915                            |

Table 29: Oracle-gap diagnostic for the guarded selector. The selected model is compared with the best candidate in the evaluated candidate set and with the best non-ensemble candidate.

| component                 | uses_source_labels | uses_unlabeled_target | uses_target_transition_labels | audit_status                                           |
|---------------------------|--------------------|-----------------------|-------------------------------|--------------------------------------------------------|
| source residual fitting   | yes                | no                    | no                            | source-city supervised residual only                   |
| target similarity weights | no                 | yes                   | no                            | obs/action, simulator output, static local descriptors |
| feature-set selector      | yes                | no                    | no                            | leave-one-source score before target evaluation        |
| leave-one-source selector | yes                | yes                   | no                            | replay matched                                         |
| ensemble dominance guard  | no                 | yes                   | no                            | max source weight threshold only                       |
| reported target metrics   | no                 | no                    | yes                           | evaluation only; not fed back to selector              |

Table 30: Target-label usage audit for the guarded source-ensemble diagnostic. Target transition labels are used only to report final metrics.

| temperature | floor | wins_vs_h2oplus | wins_vs_unweighted_cfcmt | mean_ratio_vs_h2oplus | mean_ratio_vs_unweighted_cfcmt |
|-------------|-------|-----------------|--------------------------|-----------------------|--------------------------------|
| 0.5         | 0.0   | 6               | 4                        | 0.6025                | 0.7449                         |
| 0.5         | 0.05  | 6               | 4                        | 0.6033                | 0.746                          |
| 0.5         | 0.1   | 6               | 4                        | 0.6041                | 0.747                          |
| 1.0         | 0.0   | 6               | 4                        | 0.6101                | 0.7554                         |
| 1.0         | 0.05  | 6               | 4                        | 0.6106                | 0.756                          |
| 1.0         | 0.1   | 6               | 4                        | 0.611                 | 0.7566                         |
| 2.0         | 0.0   | 6               | 4                        | 0.6145                | 0.7614                         |
| 2.0         | 0.05  | 6               | 4                        | 0.6147                | 0.7617                         |
| 2.0         | 0.1   | 6               | 4                        | 0.6149                | 0.762                          |

Table 31: Source-weighting sensitivity over temperature and floor.

| source_size | splits | cfcmt_wins_vs_h2oplus | mean_ratio_vs_h2oplus |
|-------------|--------|-----------------------|-----------------------|
| 1           | 12     | 6                     | 0.9272                |
| 2           | 12     | 8                     | 0.8705                |
| 3           | 4      | 4                     | 0.7587                |

Table 32: Mean cross-city ratio by number of source cities.

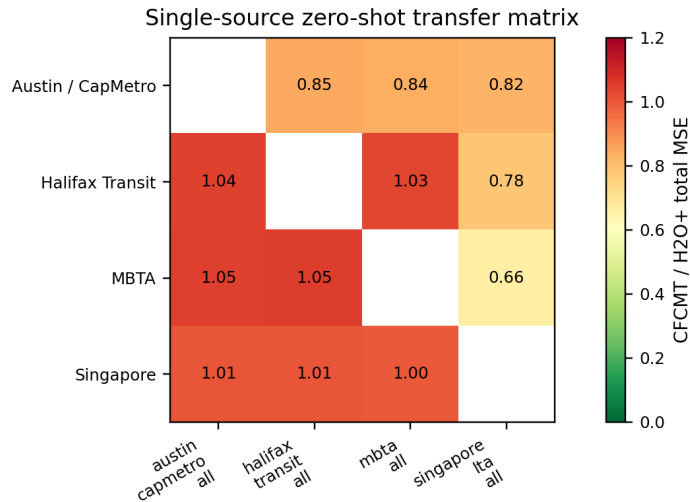


Figure 11: Single-source zero-shot transfer matrix. Cells report CFCMT/H2O total-MSE ratio with no target transition labels before reporting.

| scenario                    | splits | weighted_wins_vs_h2oplus | unweighted_wins_vs_h2oplus | weighted_wins_vs_unweighted | mean_unweighted_ratio_vs_h2oplus | mean_weighted_ratio_vs_h2oplus | mean_weighted_ratio_vs_unweighted |
|-----------------------------|--------|--------------------------|----------------------------|-----------------------------|----------------------------------|--------------------------------|-----------------------------------|
| baseline                    | 4      | 4                        | 4                          | 3                           | 0.7587                           | 0.5782                         | 0.7329                            |
| headway_gap_bias_125        | 4      | 3                        | 3                          | 3                           | 0.7587                           | 0.5768                         | 0.7301                            |
| demand_stop_reward_bias_125 | 4      | 4                        | 4                          | 3                           | 0.7587                           | 0.5802                         | 0.737                             |
| speed_bias_150              | 4      | 4                        | 4                          | 3                           | 0.7589                           | 0.5786                         | 0.7332                            |
| mixed_mechanism_bias        | 4      | 3                        | 3                          | 3                           | 0.7588                           | 0.5767                         | 0.7295                            |
| measurement_noise_10pct     | 4      | 4                        | 4                          | 3                           | 0.778                            | 0.614                          | 0.7648                            |
| measurement_noise_25pct     | 4      | 4                        | 4                          | 3                           | 0.8391                           | 0.7255                         | 0.851                             |

Table 33: Generator-bias and measurement-noise robustness.

| subset                                 | cities                                              | splits | weighted_wins_vs_h2oplus | unweighted_wins_vs_h2oplus | mean_unweighted_ratio_vs_h2oplus | mean_weighted_ratio_vs_h2oplus | mean_weighted_ratio_vs_unweighted |
|----------------------------------------|-----------------------------------------------------|--------|--------------------------|----------------------------|----------------------------------|--------------------------------|-----------------------------------|
| all_four_strict_leave_one_out          | Singapore, Austin / CapMetro, Halifax Transit, MBTA | 4      | 4                        | 4                          | 0.7587                           | 0.5782                         | 0.7329                            |
| exclude_singapore_gtfs_traffic_only    | Austin / CapMetro, Halifax Transit, MBTA            | 3      | 1                        | 1                          | 1.0025                           | 0.9984                         | 0.9964                            |
| exclude_austin_schedule_demand_proxy   | Singapore, Halifax Transit, MBTA                    | 3      | 3                        | 3                          | 0.7588                           | 0.6636                         | 0.8124                            |
| north_american_observed_ridership_pair | Halifax Transit, MBTA                               | 2      | 0                        | 0                          | 1.0421                           | 1.0421                         | 1.0                               |

Table 34: Source-subset robustness.

| policy                            | episodes | mean_reward | city_reward_gain | mean_headway_abs_error | city_headway_ratio | city_bunching_ratio | hold_seconds |
|-----------------------------------|----------|-------------|------------------|------------------------|--------------------|---------------------|--------------|
| no_hold                           | 80       | -681.1014   | 0.0              | 472.9819               | 1.1801             | 1.4256              | 0.0          |
| fixed_30                          | 80       | -709.0151   | -27.9138         | 482.1026               | 1.1852             | 1.0873              | 28.125       |
| threshold_equalization_policy     | 80       | -679.4363   | 1.6651           | 468.8503               | 1.134              | 1.2468              | 1.1384       |
| h2oplus_dense_policy              | 80       | -615.8469   | 65.2545          | 440.0112               | 1.0                | 1.0                 | 12.2202      |
| cfcmnt_mechanism_policy           | 80       | -642.9006   | 38.2008          | 451.4064               | 0.9752             | 0.9688              | 5.3885       |
| cfcmnt_similarity_weighted_policy | 80       | -644.8675   | 36.2339          | 448.0122               | 0.9601             | 0.9786              | 6.1688       |

Table 35: Expanded auxiliary sampled BusSimEnv rollout over runnable line environments from all four city bundles. City-normalized columns are included because raw rewards have different city scales.

| policy                            | paired_lines | city_reward_delta | reward_delta_c95    | city_headway_ratio | headway_ratio_c95 | city_bunching_ratio | bunching_ratio_c95 | reward_city_wins | headway_city_wins | bunching_city_wins |
|-----------------------------------|--------------|-------------------|---------------------|--------------------|-------------------|---------------------|--------------------|------------------|-------------------|--------------------|
| cfcmnt_mechanism_policy           | 80           | -27.0537          | [-68.210, 11.355]   | 0.9752             | [0.919, 1.020]    | 0.9688              | [0.877, 1.044]     | 2                | 3                 | 3                  |
| cfcmnt_similarity_weighted_policy | 80           | -29.0206          | [-74.282, 11.014]   | 0.9601             | [0.898, 1.008]    | 0.9786              | [0.877, 1.052]     | 3                | 3                 | 3                  |
| fixed_30                          | 80           | -93.1682          | [-156.302, -33.843] | 1.1852             | [1.125, 1.262]    | 1.0873              | [0.946, 1.694]     | 0                | 0                 | 1                  |
| no_hold                           | 80           | -65.2545          | [-107.566, -29.740] | 1.1801             | [1.128, 1.241]    | 1.4256              | [1.171, 2.557]     | 1                | 0                 | 0                  |
| threshold_equalization_policy     | 80           | -63.5894          | [-105.466, -27.676] | 1.134              | [1.090, 1.183]    | 1.2468              | [1.094, 1.901]     | 1                | 1                 | 1                  |

Table 36: City-stratified paired bootstrap for the auxiliary sampled rollout. Each row compares the same city-line episodes against the dense H2O<sup>+</sup>-style policy; reward deltas above zero and ratios below one are favorable.

| policy                                | cities | lines | events | handled_stop_events | warmup_first_arrivals | mean_reward | mean_headway_abs_error | bunching_rate | large_gap_rate | hold_seconds | dwell_seconds |
|---------------------------------------|--------|-------|--------|---------------------|-----------------------|-------------|------------------------|---------------|----------------|--------------|---------------|
| no_hold                               | 4      | 241   | 10000  | 17102               | 7102                  | -683.8434   | 683.8434               | 0.4924        | 0.0718         | 0.0          | 8.0           |
| daganzo_a08_policy                    | 4      | 261   | 9550   | 17531               | 7981                  | -531.0129   | 529.6356               | 0.3432        | 0.0762         | 27.5459      | 35.5459       |
| target_summary_daganzo_policy         | 4      | 261   | 9583   | 17561               | 7978                  | -529.9848   | 528.6339               | 0.3403        | 0.0763         | 27.0175      | 35.0175       |
| threshold_equalization_policy         | 4      | 252   | 9574   | 17521               | 7947                  | -532.6508   | 531.3493               | 0.3356        | 0.075          | 26.0301      | 34.0301       |
| h2oplus_dense_mpc_policy              | 4      | 241   | 9830   | 16838               | 7008                  | -685.6597   | 685.554                | 0.4875        | 0.0679         | 2.1128       | 10.1128       |
| cfcmnt_similarity_weighted_mpc_policy | 4      | 243   | 9633   | 16854               | 7221                  | -645.939    | 645.3518               | 0.4432        | 0.0653         | 11.7443      | 19.7443       |

Table 37: Libsumo full-line closed-loop holding diagnostic after adding explicit generated bus-stop dwell points. Learned residual policies use the same {0, 15, 30, 45, 60} second hold grid; the target-summary Daganzo prior uses only static route/timetable summaries to gate a classical holding gain. First-arrival warm-up events are excluded from metrics. The table validates that policy actions alter executable SUMO stop durations, but it is not a field-calibrated counterfactual experiment.

| policy                                | city_count | paired_lines | city_reward_delta | reward_delta_c95   | city_headway_ratio | headway_ratio_c95 | city_bunching_ratio | bunching_ratio_c95 | reward_city_wins | headway_city_wins | bunching_city_wins |
|---------------------------------------|------------|--------------|-------------------|--------------------|--------------------|-------------------|---------------------|--------------------|------------------|-------------------|--------------------|
| cfcmnt_similarity_weighted_mpc_policy | 4          | 241          | 39.5999           | [29.938, 51.745]   | 0.8641             | [0.819, 0.902]    | 0.8482              | [0.792, 0.896]     | 3                | 3                 | 3                  |
| daganzo_a08_policy                    | 4          | 241          | 154.7234          | [133.683, 184.068] | 0.6715             | [0.635, 0.704]    | 0.6544              | [0.596, 0.710]     | 4                | 4                 | 3                  |
| no_hold                               | 4          | 241          | 1.8163            | [-6.305, 8.868]    | 0.8532             | [0.837, 0.877]    | 1.0129              | [0.993, 1.036]     | 3                | 3                 | 0                  |
| target_summary_daganzo_policy         | 4          | 241          | 155.7531          | [133.138, 184.261] | 0.6595             | [0.625, 0.692]    | 0.6496              | [0.593, 0.698]     | 4                | 4                 | 3                  |
| threshold_equalization_policy         | 4          | 241          | 153.6737          | [133.529, 181.908] | 0.6809             | [0.645, 0.714]    | 0.6384              | [0.584, 0.689]     | 4                | 4                 | 3                  |

Table 38: City-stratified paired line bootstrap for the libsumo full-line holding diagnostic. Rows compare city-line policy summaries against dense H2O<sup>+</sup>-MPC when the one-step dense selector is absent; reward deltas above zero and ratios below one are favorable.



| Algorithm | Scenario     | Runs | Time loss | Duration | Waiting | Elapsed (s) |
|-----------|--------------|------|-----------|----------|---------|-------------|
| IDQN      | arterial4x4  | 3    | 1505.536  | 501.305  | 436.920 | 17179.205   |
| MPLight   | arterial4x4  | 3    | 1056.797  | 357.454  | 238.745 | 13576.024   |
| IDQN      | cologne1     | 3    | 612.737   | 164.905  | 138.950 | 812.441     |
| MPLight   | cologne1     | 3    | 189.393   | 122.751  | 68.843  | 6985.784    |
| IDQN      | cologne3     | 3    | 86.610    | 92.313   | 43.268  | 2351.500    |
| MPLight   | cologne3     | 3    | 1011.239  | 543.511  | 515.329 | 10132.753   |
| IDQN      | cologne8     | 3    | 24.760    | 89.379   | 8.354   | 5946.643    |
| IDQN      | grid4x4      | 3    | 47.752    | 158.215  | 25.205  | 11120.623   |
| MPLight   | grid4x4      | 3    | 44.393    | 156.112  | 16.598  | 10392.985   |
| IDQN      | ingolstadt1  | 3    | 18.097    | 37.191   | 5.490   | 911.332     |
| MPLight   | ingolstadt1  | 3    | 401.895   | 85.906   | 63.192  | 7888.143    |
| IDQN      | ingolstadt21 | 3    | 256.588   | 353.111  | 176.405 | 13141.256   |
| IDQN      | ingolstadt7  | 3    | 51.869    | 78.827   | 17.357  | 5313.646    |
| MPLight   | ingolstadt7  | 3    | 284.391   | 111.560  | 57.463  | 8755.139    |

Table 39: Official RESCO full-budget RL runner results under the upstream native protocol. Rows are not same-protocol comparisons against CFCMT; they use RESCO’s native observation, reward, action, training, and evaluation definitions.

| Agent      | Network | Runs | Travel time | Queue  | Delay | Throughput | Elapsed (s) |
|------------|---------|------|-------------|--------|-------|------------|-------------|
| dqn        | sumo1x1 | 3    | 38.945      | 3.240  | 0.283 | 1998.667   | 175.967     |
| frap       | sumo1x1 | 3    | 41.923      | 24.461 | 0.460 | 1664.667   | 439.620     |
| mplight    | sumo1x1 | 3    | 40.546      | 4.373  | 0.319 | 1995.000   | 371.569     |
| presslight | sumo1x1 | 3    | 39.748      | 3.442  | 0.285 | 1998.667   | 175.835     |
| dqn        | sumo1x3 | 3    | 97.090      | 6.531  | 0.244 | 2487.333   | 425.384     |
| frap       | sumo1x3 | 3    | 118.712     | 11.672 | 0.398 | 2505.333   | 1583.053    |
| mplight    | sumo1x3 | 3    | 66.502      | 1.876  | 0.243 | 2814.333   | 647.664     |
| presslight | sumo1x3 | 3    | 63.188      | 1.164  | 0.188 | 2816.000   | 393.414     |
| dqn        | sumo4x4 | 3    | 140.610     | 0.603  | 0.041 | 1465.000   | 2140.773    |
| frap       | sumo4x4 | 3    | 140.477     | 0.612  | 0.042 | 1464.000   | 7188.185    |
| mplight    | sumo4x4 | 3    | 140.469     | 0.607  | 0.042 | 1464.333   | 1017.297    |
| presslight | sumo4x4 | 3    | 144.279     | 0.711  | 0.048 | 1463.333   | 2080.043    |

Table 40: Patched LibSignal full-budget RL runner results under the native LibSignal protocol. Rows are included as a reviewer-facing RL baseline audit, not as direct phase-MPC wins or losses against CFCMT.

| Policy                  | Mean queue | Seed std | P90 queue | Trip time | Trip wait | Time loss | Throughput |
|-------------------------|------------|----------|-----------|-----------|-----------|-----------|------------|
| Phase pressure          | 32.2343    | 0.9901   | 62.1800   | 104.2291  | 20.8528   | 38.5434   | 0.9492     |
| MaxPressure             | 42.7872    | 0.7314   | 71.6277   | 106.9182  | 20.5521   | 40.7331   | 0.9448     |
| CFCMT + pressure guard  | 43.5106    | 0.5699   | 74.2496   | 108.6326  | 22.0400   | 42.4485   | 0.9448     |
| Target-static simulator | 44.8973    | 1.2335   | 65.4094   | 258.2547  | 169.8900  | 188.9974  | 0.8754     |
| H2O+-style dense        | 49.5058    | 3.9510   | 74.9880   | 221.4173  | 131.1706  | 150.3275  | 0.8827     |
| Native actuated SUMO    | 58.3468    | 1.1370   | 99.4055   | 164.5754  | 66.1326   | 98.1604   | 0.9714     |
| Fixed RESCO program     | 84.6088    | 0.6458   | 133.0312  | 221.2093  | 113.3988  | 154.3776  | 0.9385     |

Table 41: Extended RESCO 3600-second long-horizon check using the same eight scenarios and three matched seeds as the main traffic-signal stress test. Trip time, trip wait, and time loss are official SUMO `tripinfo` metrics over completed vehicles and should be read together with throughput; lower queue and higher throughput are primary.

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